

MINISTRY OF EDUCATION & TRAINING

STATE BANK OF VIETNAM

**HO CHI MINH UNIVERSITY OF BANKING**



**BACHELOR THESIS**

**FACTORS AFFECTING THE STOCK PRICE OF JOINT-STOCK  
COMMERCIAL BANKS IN VIETNAM**

MAJOR : FINANCE – BANKING  
CODE : 7340201

NAME : NGUYEN DANG MINH HUNG  
ID : 050609211951  
CLASS : HQ9-GE21  
SUPERVISOR : PhD. NGO VAN TUAN

**HO CHI MINH CITY, 2025**

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**HO CHI MINH CITY, 2025**

**COMMENTS FROM THE THESIS INSTRUCTOR**

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Ho Chi Minh, date...,month..., 2024

. **Supervisor**

PhD. Ngo Van Tuan

**REVIEW BOARD COMMENTS**

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Ho Chi Minh, date...,month..., 2024

REVIEW BOARD COMMENTS

## **ABSTRACT**

This study will clarify and analyze the macro factors and endogenous factors from within the bank and show how the relationship and connection between the factors will affect stock prices with data taken from official sources in the period 2013-2024 by estimating through appropriate regression models such as Pooled OLS, FEM, REM, GLS. The thesis shows that it will bring much helpful information to readers. In addition, this study also assesses the impact of some macro and internal bank factors such as bank size ratio (SIZE), cost-to-income ratio (CIR), dividend payout ratio (DPS), growth rate of domestic product (GDP), and inflation rate (INF). Based on the analysis results, the study proposes some policy implications to support the system of commercial banks and supervisory agencies, improve financial capacity, and promote sustainable development in the banking sector.

## **COMMITMENT**

My name is Nguyen Dang Minh Hung, a student of class HQ9-GE21, student code: 050609211951, majoring in Finance - Banking at Banking University of Ho Chi Minh City.

I hereby declare that the thesis with the topic "Factors affecting the stock prices of commercial banks in Vietnam" is a research paper directly conducted by me. All content, including data, research methods, analysis and conclusions, are conducted transparently, not copied from any other research, except for the content that has been fully cited and sourced according to regulations.

I pledge that this research does not violate the regulations on plagiarism or any form of academic fraud. If there are any errors or violations related to the honesty of the thesis, I will take full responsibility and accept all forms of handling according to the regulations of the school.

I sincerely pledge!

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## **CHAPTER 1: INTRODUCTION TO THE TOPIC**

### **1.1 PROBLEM STATEMENT AND URGENCY OF THE TOPIC**

The urgency of the topic is determined based on the following specific factors, reflecting current challenges and practical needs:

2020-2024 had written a tumultuous chapter for the Vietnamese economy after COVID-19, when the country faced profound changes after more than a decade since 2013. The General Statistics Office of Vietnam (2024) recorded a GDP growth of 6.5% in 2022. However, it decreased to 4.8% in 2023 due to global inflation and supply chain disruptions, causing credit demand to decline and affecting the profits and stock prices of banks. In this context, the pressure to manage non-performing loans (NPLs) has become more evident, with the State Bank of Vietnam (2023) reporting that NPLs will reach 1.8% of total outstanding loans by the end of 2023. However, some smaller banks still face higher rates, undermining investor confidence – a problem Barth et al. (2019) warned about through research in 180 countries. At the same time, the Fintech and digital transformation wave has breathed new life into the banking industry, with names such as Techcombank, TPBank, and MB Bank reducing their cost-to-income ratio (CIR) from 50% to around 40% in 2020-2023 (State Bank of Vietnam, 2023). However, Vives (2019) notes that this competitive pressure also brings opportunities and challenges to stock prices. In addition, the international economic integration process from 2013-2024 has opened up opportunities with the foreign ownership ratio limited to 30% under the Investment Law 2020, when funds such as Warburg Pincus invested in Techcombank and VPBank, bringing advanced governance and raising stock prices (Nguyen & Tran, 2020). However, Claessens and Van Horen (2014) emphasized that this limit poses challenges that must be explored. These fluctuations

not only pose practical needs from investors – needing information on EPS, BM, and Foreign Ownership – but also from banks – wanting to improve CIR, NPL, DPS – and policymakers – need data to adjust GDP, INF, and Vnibor, as Ross et al. (2020) suggested. However, previous studies in Vietnam, such as Le & Nguyen (2021) and Pham & Ho (2022), mainly focused on individual factors (GDP, EPS, NPL) based on data up to 2019, ignoring the comprehensive picture in the post-pandemic period. This raises the urgent need for a new research model, integrating macro and internal factors, to not only decipher the impact on bank stock prices but also support stakeholders in overcoming future economic challenges. Globally, many studies have explored the impact of economic crises on financial markets, but most of them focused on developed markets or general aspects. In Vietnam, some works, such as Nguyen and Tran (2020), analyzed the factors affecting bank stock prices in 2015-2019, emphasizing the role of GDP growth, bank size, and profit (EPS). However, these studies have not fully covered the impact of the COVID-19 pandemic, especially in the long-term context from 2013-2024. This period witnessed not only the shock from the pandemic but also changes in monetary policy, foreign ownership ratio, and pressure from bad debt, affecting stock prices. For example, from 2020-2021, when operating interest rates increased to control inflation, the net interest margin (NIM) of banks decreased, while 2023-2024 recorded a recovery thanks to foreign investment inflows into large banks such as Techcombank. These factors, along with dividend policy (DPS) and cost-to-income ratio (CIR), remain gaps that need to be further studied. For the above reasons, the topic "Factors affecting stock prices of Vietnamese commercial banks" was born to fill this gap. The study will analyze how macroeconomic factors (GDP, inflation), bank-specific factors (size, EPS, DPS, NPL, CIR), and foreign ownership ratio affect stock prices during 2013-2024, focusing on post-COVID-19 volatility. By using real-world data from listed banks such as Vietcombank, BIDV, and Techcombank, the study not only provides a comprehensive view of how the pandemic

and other factors shape the Vietnamese stock market but also supports bank management in adjusting stock pricing strategies, helping investors make wise decisions, and providing a basis for policymakers in formulating future financial stability measures.

## **1.2 OBJECTIVES OF THE SUBJECT**

### **1.2.1 General objectives**

The general objective of the topic is to analyze and evaluate the factors affecting the stock prices of commercial banks in Vietnam based on a regression model integrating macroeconomic **factors (GDP, INF)**, internal factors of the bank (**SIZE, EPS, BM, DPS, NPL, CIR**), and **foreign ownership ratio (Foreign Ownership)**. This paper aims to provide a comprehensive view on the valuation of stock prices, thereby providing practical policy implications that can enhance the valuation of stock prices of commercial banks in Vietnam in the current context.

### **1.2.2 Specific Objectives**

To achieve the general objective, the topic sets out the following specific objectives:

The general objective of this study is to identify and evaluate the impact of different factors on the stock value of the system of joint stock commercial banks in Vietnam.

To complete the general objective, this study sets out three specific objectives:

1. First, identify the factors affecting the stock price value of joint stock commercial banks in Vietnam

2. Second, review and analyze the impact of macroeconomic factors and internal factors affecting the stock price value of joint commercial banks in Vietnam.
3. Third, based on quantitative analysis, propose some policy implications to improve the stock price value of joint stock commercial banks under the current economic conditions in our country.

### **1.3 RESEARCH QUESTIONS**

To achieve the above objectives, the topic raises the following research questions:

1. Firstly, what factors affect the stock prices of joint stock commercial banks in Vietnam?
2. Secondly, do those factors significantly affect the stock prices of banks? Moreover, are those impacts positive or negative?
3. Thirdly, what policy implications and practical solutions can be proposed to enhance and improve the stock prices of joint stock commercial banks in Vietnam in the current economic context and competitive pressure?

### **1.4 SUBJECTS AND SCOPE OF THE RESEARCH**

#### **1.4.1 Research subjects**

The research subjects in the research article "Factors affecting the stock prices of commercial banks in Vietnam," specifically in the period 2013-2024, mainly focus on deepening and researching new factors on macro and micro factors of the market, in which it is impossible not to mention GDP (economic growth rate), factors related to INF (inflation), in addition to many other macro factors that have a profound impact on the economic context. Internal factors of the bank SIZE (scale), EPS (earnings per share), BM (book value to market price ratio), DPS (dividend payout ratio), NPL (bad

debt ratio), and CIR (cost to income ratio) as financial and governance indicators. Foreign ownership ratio (Foreign Ownership). Considered as a market factor, reflecting the level of international integration and governance.

#### **1.4.2 Research scope**

Time scope: The study was conducted from 2013-2024, reflecting the fluctuating period of the Vietnamese stock market. From stability after the 2008 financial crisis (VN-Index 400-500 points), growth to 2019 (900 points), suffering the COVID-19 shock (2020-2022) with increasing bad debt and decreasing credit demand, to recovery in 2023-2024 thanks to flexible monetary policy and foreign investment. This scope allows analysis of the long-term impact of macroeconomic factors (GDP, inflation) and internal factors (EPS, DPS, NPL, CIR, foreign ownership) on bank stock prices, meeting the needs of practical research.

Spatial scope: The study focuses on the joint stock commercial banking industry in Vietnam, with banks listed on HOSE and HNX, reflecting financial health and market confidence. The banking industry is the economic backbone, supporting credit, regulating the macro economy, and promoting growth in the context of Vietnam's integration from 2013-2024. The subjects include large banks (Vietcombank, BIDV, Techcombank) and small-medium banks (Sacombank, MB Bank), representing diversity in governance and international integration, bad debt pressure, providing a comprehensive view, supporting investors, bank managers, and policymakers in the context of rapid economic transformation in 2024.

### **1.5 CONTRIBUTIONS OF THE TOPIC**

In this study, the author expects to bring readers more theoretical contributions, providing a regression model integrating additional macroeconomic factors (GDP,

INF) and internal factors of the banking industry: SIZE, BM, CIR, NPL, and DPS. Particularly worth mentioning is the foreign ownership ratio (Foreign Ownership), supplementing previous studies such as Nguyen & Tran (2020) and Pham & Ho (2022). In addition, it also provides a scientific basis for investors to evaluate stocks based on macro and internal aspects to improve the current situation at commercial banks and provide more information for macroeconomic policymakers (Ross et al., 2020). Using data from 2013-2024 reflects the context before and after the COVID-19 pandemic and the digital explosion, overcoming the limitations of previous studies such as Le & Nguyen (2021). Applying advanced regression models, combined with quantitative analysis, to analyze influencing factors, expanding the approach of studies such as Fama & French (1993).

## **1.6 THESIS STRUCTURE**

The thesis is structured with five main chapters

**Chapter 1:** Introduction to the topic (current page): Problem statement, defining objectives, scope, methods, and contributions of the research.

**Chapter 2:** Theoretical basis theory and research overview: Present relevant theories, influencing factors, and a review of domestic and foreign research to serve as a foundation for analysis.

**Chapter 3:** Research methodology: Detail the regression model, data sources (expected from financial reports and market data 2013-2024), and analysis process.

**Chapter 4:** Analysis and Evaluation: Includes the implementation of quantitative methods along with the evaluation of other factors affecting bank stock prices.

**Chapter 5:** Conclusion and Policy Implications: Summarizes the new findings, along with suggestions for implications and future research directions that are useful

## **CHAPTER 2: THEORETICAL BASIS AND OVERVIEW OF PREVIOUS RESEARCH**

### **2.1 THEORETICAL BASIS**

#### **2.1.1 Concept of shares**

Shares are a type of security that confirms the investor's ownership of a part of the charter capital of the joint stock company that issues them. According to Ross et al. (2020), shares represent the economic rights of shareholders, including the right to receive dividends and the right to vote in important decisions of the company at the general meeting of shareholders. Ordinary shares are the most common type, especially in the banking industry, where commercial banks issue shares to raise capital on the HOSE or HNX.

In the banking industry, shares are not only a tool for raising capital but also a measure reflecting the financial health and market confidence in the bank. The theory of bank stock characteristics indicates that bank stock prices in Vietnam are often more volatile than other industries due to their sensitivity to macroeconomic factors such as GDP, inflation, and interest rates, as well as internal factors such as non-performing loan (NPL) ratio and earnings per share (EPS). Moreover, bank stocks also play an important role in supporting credit and investment activities, promoting economic growth (Le & Nguyen, 2021).

#### **2.1.2 Types of Common Stock Prices**

Common stock values are classified quite diversely based on the method of determination and the market context, usually including about four main types: nominal price, market price, book value, and intrinsic value. Par Value, a fixed price per share, has a legal and symbolic nature. Although it does not reflect the actual value or market expectations, it plays an important role in legal transactions such as issuing

new shares. Market Price is the actual trading price on the stock exchange, affected by supply and demand, based on financial information and measured by the average annual closing price on HOSE or HNX (Nguyen & Tran, 2020). This is the type of price used in this study. Book Value is the net asset value per share, calculated by dividing equity by the number of outstanding shares. This is an important basis for calculating the book value to market value ratio (B/M), a variable in the Fama-French model (1993). In Vietnam, large banks often have high book values thanks to their large asset size and strong capital foundation. Intrinsic Value is an estimated price based on expected future cash flows discounted to the present. According to Damodaran (2012), the intrinsic value depends on expected profits, dividends, growth rate, and risk level. The difference between intrinsic value and market price is the basis for investors to buy or sell stocks. In the context of Vietnam, bank stock prices are often affected by factors such as dividend payout ratio (DPS) and non-performing loan ratio (NPL) (Tran & Le, 2017). In addition, the development of Fintech and international integration have increased stock price volatility. Factors such as foreign ownership ratio (Foreign Ownership) and cost management efficiency (CIR) are increasingly of interest to investors (Vives, 2019).

### **2.1.3 Efficient Market Hypothesis**

The Efficient Market Hypothesis (EMH) developed by Fama (1970) states that stock prices fully reflect all available information in the market. According to this theory, EMH is divided into three primary forms. The first is the Weak Form, in which stock prices have reflected all historical information, such as past prices and trading volumes. This means that technical analysis cannot provide an advantage in predicting stock prices. The second is the Semi-Strong Form, when stock prices reflect all public information, including financial statements, macroeconomic data such as GDP and inflation, and internal information of the business such as EPS or NPL. Finally, there is

the Strong Form, where stock prices reflect unpublished inside information, making it impossible for insiders to achieve excess returns.

For the banking industry, EMH implies that factors such as GDP, CPI, or EPS will be immediately reflected in stock prices by the market.

#### **2.1.4 The Random Walk Theory**

The Random Walk Theory, proposed by Malkiel (1973), argues that stock price movements are random and cannot be predicted based on historical information. This theory is closely related to the Weak Form of EMH, which asserts that stock prices move randomly due to new information being reflected immediately, without any pattern. Malkiel (1973) argues that investors cannot “beat the market” using technical analysis or strategies based on historical data.

Stock prices tend to fluctuate randomly, especially during periods of high market volatility (such as the 2008-2009 financial crisis or the COVID-19 pandemic). However, over the long term, the fundamental effects of factors such as EPS, NPL, CIR, and Foreign Ownership are significant, suggesting that the random walk theory does not apply in all cases.

#### **2.1.5 The Signalling Theory**

The Signalling Theory, developed by Spence (1973), argues that corporate actions send signals to investors about prospects. In the banking industry, important signals include the announcement of earnings per share (EPS), dividend payments (DPS), and management decisions such as increasing foreign ownership. Lintner (1956) demonstrates that increasing dividends sends a positive signal about stable profitability while decreasing dividends can signal financial distress. Easterbrook (1984) adds that high dividends help reduce agency conflicts between shareholders and management, thereby increasing investor confidence.

Meanwhile, dividend policy (DPS) to send signals about financial stability attracts long-term investors. Similarly, the announcement of high EPS is a strong signal about operating efficiency, positively affecting stock prices (Pham & Ho, 2022). This theory will be used in the study to explain the impact of factors such as EPS, DPS, and CIR on bank stock prices.

## **2.2 Factors Affecting Stock Market Value**

### **2.2.1 Group of macro factors**

#### **2.2.1.1 Gross Domestic Product (GDP)**

Gross domestic product (GDP) is a basic macroeconomic indicator, reflecting the total value of final goods and services produced in a country. According to the efficient market theory (Fama, 1970), GDP is public information quickly reflected in stock prices, especially in the banking industry, where economic growth drives credit demand and profits. High GDP growth rates indicate a growing economy, encouraging commercial banks to expand their scale and increase revenue and profits, positively affecting stock prices. On the contrary, low GDP growth signals a stagnant economy, causing investors to withdraw capital and putting downward pressure on stock prices. Research by Pham & Ho (2022) in Vietnam shows that GDP has a positive (+) relationship with bank stock prices. GDP growth is often accompanied by increased demand for loans from businesses and individuals, increasing bank revenue and profits, creating a positive signal for the market. This result is consistent with the signaling theory (Spence, 1973), when high GDP sends signals about positive economic prospects, boosting investor confidence. However, during volatile periods such as COVID-19 (2020-2022), the impact of GDP may be reduced due to other factors such as bad debt.

#### **2.2.1.2 Inflation (INFLATION)**

Inflation is an increase in the price of goods and services over time compared to the previous period. According to the efficient market theory (Fama, 1970), inflation is

public information reflected in stock prices by the market. However, the impact of inflation on bank stock prices may vary depending on the context. Research by Pham & Ho (2022) in Vietnam shows that inflation has no clear statistical significance on bank stock prices, consistent with the results from random walk theory (Malkiel, 1973) when stock prices fluctuate randomly and are difficult to predict based on inflation alone.

According to signaling theory (Spence, 1973), high inflation can send negative signals about increased operating costs, reducing bank profits and negatively affecting stock prices. Conversely, in some cases, moderate inflation can stimulate loan demand, increasing bank revenue. However, in Vietnam, studies such as Pham and Ho (2022) show that the impact of inflation is often unclear in the short term, especially when internal factors such as EPS or NPL have a more substantial influence.

## **2.2.2 Group of endogenous factors within banks**

### **2.2.2.1 Bank size (SIZE)**

Bank size, measured by total assets, equity, or deposits, is an important endogenous factor affecting stock prices. According to signaling theory (Spence, 1973), large scale sends signals about the competitiveness and reliability of the bank, attracting investors. Large banks such as Vietcombank or BIDV often have high liquidity, invest in modern technology and professional management systems, and improve operational efficiency and market confidence.

Research by Pham & Ho (2022) in Vietnam confirms that bank size has a positive (+) relationship with stock prices. This is consistent with the efficient market theory (Fama, 1970), when size information is quickly reflected in stock prices. Large banks often have advantages in providing diverse services, reducing transaction costs, and

maintaining a competitive position, thereby increasing stock value. However, in times of crisis, large size can also increase systemic risk, which needs to be carefully considered.

#### **2.2.2.2 Earnings per share (EPS)**

Earnings per share (EPS) is an important indicator measuring the profit allocated to each share, reflecting the operational efficiency of the bank. According to the signaling theory (Spence, 1973), high EPS sends a positive signal about profitability, attracts investors, and increases stock prices. EPS is calculated by dividing after-tax profit by the number of outstanding shares, an indicator that investors often consider when evaluating the financial health of banks.

Research by Pham & Ho (2022) in Vietnam noted a positive (+) relationship between EPS and bank stock prices, consistent with the efficient market theory (Fama, 1970) when information about EPS is quickly reflected by the market. High EPS shows that the bank operates efficiently, increasing investors' profit expectations and boosting stock prices. Conversely, low EPS can signal financial difficulties, reducing confidence and stock prices.

#### **2.2.2.3 Book-to-Market Ratio (BM)**

The book-to-market ratio (BM) measures the relationship between the accounting value and the market value of a stock, calculated as equity divided by market capitalization. According to the efficient market theory (Fama, 1970), BM reflects public information about the intrinsic value of a bank. A high BM suggests that the stock may be undervalued, while a low BM reflects high growth expectations from the market.

A study by Pham & Ho (2022) in Vietnam shows a negative relationship (-) between BM and bank stock prices. This is consistent with the signaling theory (Spence, 1973), when a low BM sends a signal about growth potential, attracting investors. Banks with

low BM, such as Vietcombank, are often valued highly due to their good performance, while high BM may signal financial risk, reducing the stock's attractiveness.

#### **2.2.2.4 Earnings per share (DPS)**

The dividend payout ratio (DPS) measures the dividends distributed per share, reflecting the bank's profit distribution policy. According to signaling theory (Lintner, 1956; Spence, 1973), high DPS sends a positive signal about financial stability, increasing investor confidence and stock prices. DPS is calculated as total dividends divided by the number of outstanding shares, an important indicator in the banking industry.

Research by Pham & Ho (2022) and Easterbrook (1984) confirms a positive (+) relationship between DPS and bank stock prices in the short term. High DPS, as at Techcombank, indicates stable profitability, attracting long-term investors. However, in the long term, reinvestment priority can reduce DPS, affecting stock prices, as Pham and Ho (2022) found.

#### **2.2.2.5 Non-Performing Loan (NPL) Ratio**

Non-Performing Loan (NPL) Ratio measures credit quality, calculated as the total non-performing loans divided by the total outstanding loans, expressed as a percentage. According to the efficient market theory (Fama, 1970), NPL is public information directly affecting stock prices. High NPL signals increased credit risk, reducing profits and investor confidence.

Pham & Ho (2022) found a negative relationship (-) between NPL and bank stock prices in Vietnam. This is consistent with the signaling theory (Spence, 1973), when low NPL signals effective risk management, increasing stock value. Banks like BIDV, with high NPL during the COVID-19 period (2020-2022), are often under pressure to reduce stock prices.

### **2.2.2.6 Cost-to-Income Ratio (CIR)**

Cost-to-income ratio (CIR) measures management efficiency, calculated by dividing total operating costs by total income, expressed as a percentage. According to the efficient market theory (Fama, 1970), CIR reflects information about operating efficiency, affecting stock prices. Low CIR indicates good cost management, increasing profits, and stock attractiveness.

Research by Pham & Ho (2022) confirms the negative relationship (-) between CIR and bank stock prices in Vietnam. This is consistent with the signaling theory (Spence, 1973); when CIR is low, as at MB Bank, thanks to digital transformation, it sends a signal about operational efficiency, attracting investors. High CIR signals poor operating costs, reducing stock prices.

### **2.2.2.7 Foreign Ownership**

Foreign ownership (FO) measures foreign investor participation, calculated by dividing the number of foreign shares held by the total shares, expressed as a percentage. According to the signaling theory (Spence, 1973), high FO sends a signal about the quality of governance and growth potential, increasing stock prices.

Research by Pham & Ho (2022) in Vietnam noted a positive (+) relationship between FO and bank stock prices. This is consistent with the efficient market theory (Fama, 1970) when information about FO is quickly reflected in stock prices. Banks like Vietcombank with high FO due to good governance and international integration often have increasing stock prices, while low FO can reduce investor confidence.

## **2.3 SUMMARY OF RELATED STUDIES**

### **2.3.1 Domestic research**

**Nguyen & Tran (2020):** Research on macroeconomic factors (GDP, CPI, Vnibor) and internal factors (SIZE, NPL, Foreign Ownership) affecting bank stock prices in Vietnam in the period 2015-2019. The results show that GDP and SIZE have a positive

relationship with stock prices, while Vnibor and NPL have a negative impact. The study also notes that Foreign Ownership has a positive impact, especially in large banks such as Vietcombank and Techcombank, where the foreign ownership ratio is close to the 30% limit. However, the study has not considered factors such as CIR and P/E, nor has it considered the impact of Fintech.

**Le & Nguyen (2021):** Focus on macroeconomic factors (GDP, CPI) and their impact on bank stock prices. The results show that GDP has a positive impact, but CPI is not statistically significant during stable inflation (2015-2019). The study also notes that large banks have scale advantages (SIZE) but has not analyzed in-depth internal factors such as EPS, B/M, and CIR. The study's data stopped before the COVID-19 pandemic, so it does not reflect the impact of recent economic fluctuations.

**Pham & Ho (2022):** Analyzing internal factors (EPS, NPL, CIR, Foreign Ownership) affecting bank stock prices in Vietnam. The results show that EPS and Foreign Ownership have positive impacts, while NPL has negative impacts. CIR is also noted as an important factor, with banks with low CIR (such as MB Bank) often having higher stock prices. This study does not mention macroeconomic factors such as GDP and CPI, and has not considered the role of Fintech in improving CIR.

**Tran & Le (2017):** Focusing on the impact of dividend policy (DPS), P/E, and EPS on bank stock prices. The results show that DPS and EPS have positive impacts, especially at banks with stable dividend strategies, such as Techcombank and VPBank. P/E is an important factor reflecting investor expectations, with banks with reasonable P/E (8–10) often attracting long-term investors. However, this study does not mention macro factors and the impact of digital transformation.

**Hoang & Nguyen (2018).** Research on the impact of NPL and CIR on bank stock prices during the restructuring period (2011-2015). The results show that high NPL reduces stock prices, especially in small banks that have not yet resolved bad debts. CIR also has a negative impact, with banks with high CIR (over 50%) often undervalued by investors. This study does not mention Foreign Ownership and macro factors such as Vnibor.

### 2.3.2 Foreign research

**Fama & French (1993).** Developing a three-factor model (size, B/M, and market risk) to explain stock price fluctuations. The study notes that B/M hurts stock prices in the financial sector, as stocks with low B/M are generally overvalued. SIZE also has a positive impact, consistent with large banks. This study does not address bank-specific factors such as NPL and CIR.

**Claessens & Van Horen (2014).** A study of the impact of Foreign Ownership on bank stock prices in 137 countries. The results show that Foreign Ownership has a positive impact, as foreign investors bring capital and advanced management experience. The study also notes that in emerging markets, this impact depends on the level of transparency and governance of the bank, a point to consider in the context of Vietnam.

**Demirgüç-Kunt & Huizinga (2019).** Analysis of CIR and internal factors in 80 countries. The results show that high CIR hurts stock prices, reflecting poor governance. The study also notes that banks with low CIR (below 40%) tend to have more stable stock prices, especially in developed markets such as the US and Europe. However, the study does not address the impact of Fintech in reducing CIR.

**Vives (2019):** Assessing the impact of Fintech and digital transformation on the banking industry in developed countries. The results show that Fintech creates competitive pressure and promotes innovation, helping banks reduce CIR and improve operational efficiency. The study suggests that factors such as CIR and DPS may indirectly reflect the impact of Fintech but have not been specifically analyzed in the context of emerging markets such as Vietnam.

**Barth et al. (2019).** Research on the impact of NPL and bank governance in 180 countries. The results show that high NPLs reduce stock prices by increasing financial risk and reducing investor confidence. The study also notes that banks with good governance (often through Foreign Ownership) are often better able to control NPLs, thereby supporting stock prices.

| No. | Author(s)<br>(Year)     | Scope                      | Main Factors                                         | Key Findings                                                                                               | Limitations                              |
|-----|-------------------------|----------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 1   | Nguyen &<br>Tran (2020) | Vietnam<br>(2015-<br>2019) | GDP, CPI, Vnibor,<br>SIZE, NPL, Foreign<br>Ownership | GDP & SIZE (+); Vnibor & NPL (-);<br>Foreign Ownership (+), especially in<br>large banks like Vietcombank, | Did not consider CIR, P/E,<br>or Fintech |

**Table 2.1: Domestic Studies**

|     |                              |                     |                                  |                                                                                                                      | Techcombank                                                         |  |
|-----|------------------------------|---------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--|
| No. | Author(s)<br>(Year)          | Scope               | Main Factors                     | Key Findings                                                                                                         | Limitations                                                         |  |
| 2   | Le & Nguyen (2021)           | Vietnam (2015-2019) | GDP, CPI                         | GDP (+); CPI (insignificant); SIZE (+)                                                                               | Did not analyze EPS, B/M, CIR, dataset ends before COVID-19         |  |
| 1   | Fama & French (1993)         | Global              | SIZE, B/M, Market Risk           | SIZE (+); B/M (-) → Lower B/M stocks often                                                                           | Did not cover banking-factors like                                  |  |
| 3   | Pham & Ho (2022)             | Vietnam             | EPS, NPL, CIR, Foreign Ownership | EPS & Foreign Ownership (+); NPL (-); Lower CIR → Higher stock prices                                                | Did not examine factors like NPG, DPI, CPI; did not address Fintech |  |
| 2   | Claessens & Van Horen (2014) | 137 countries       | Foreign Ownership                | Foreign Ownership (+); Depends on DPS & EPS (& reasonable P/E (8-10) → Attracts long-term investors                  | Requires careful application to economic                            |  |
| 4   | Demirgüç-Kunt (2017)         | Vietnam             | DPS, P/E, EPS                    | Higher CIR → Negative impact; Lower CIR → More stable stock prices, especially in developed markets like the US & EU | Ignored Fintech's impact; did not cover digital transformation      |  |
| 3   | Edling & Raga (2019)         | 80 countries        | CIR, Internal factors            | Higher NPL & CIR → Lower stock prices; Particularly affects smaller banks                                            | Did not examine Fintech's impact                                    |  |
| 5   | Nguyen (2018)                | Vietnam (2011-2015) | NPL, CIR                         | Fintech → Innovation, reduces CIR, improves factors                                                                  | Did not include Foreign Ownership; ignored macro factors            |  |
| 4   | Ves (2019)                   | Developed           | Fintech, CIR,                    | Fintech → Innovation, reduces CIR, improves factors                                                                  | Not specific to                                                     |  |

|   |                     |               |                 |                                                                                                                            |                               |
|---|---------------------|---------------|-----------------|----------------------------------------------------------------------------------------------------------------------------|-------------------------------|
|   |                     | countries     | DPS             | efficiency; CIR & DPS indirectly reflect Fintech impact                                                                    | emerging markets like Vietnam |
| 5 | Barth et al. (2019) | 180 countries | NPL, Governance | Higher NPL → Lower stock prices; Good governance (often via Foreign Ownership) → Better NPL control → Supports stock price |                               |

**Table 2.2: International Studies**

### 2.3.3 Research gaps and novelty of the research topic

On a theoretical basis, the author stated that nine factors affect the liquidity of commercial banks. In 8 related research articles, there are almost all nine factors that affect the research article, and many other factors appear. Most of the related research articles at home and abroad have research gaps that show that the asset size, the ratio of operating expenses to income, the ratio of bad debt, the GDP growth rate, and inflation have a profound impact on the stock price value of commercial banks on the stock market. However, few studies combine many macroeconomic factors and internal factors of the bank to affect the value of shares of commercial banks. In addition, most of the previous studies used old data, so the author decided to study this topic with new variables such as *foreign ownership ratio* and cost to *operating income ratio (CIR)* and update all the latest data to get the most comprehensive and latest research results.

## SUMMARY OF CHAPTER 2

Chapter 2 provided the theoretical basis and overview of studies on factors affecting bank stock prices. Based on theories such as EMH, Random Walk Theory, and Signalling Theory, the study identified macroeconomic factors (GDP, CPI) and internal factors of the bank (SIZE, EPS, P/E, B/M, NPL, CIR, Foreign Ownership) as the main variables. The review of previous research shows diversity in approaches, but there are still many gaps, especially in multidimensional integration using updated data. With research directions such as comprehensive models, post-COVID-19 data, and a focus on Foreign Ownership, the thesis promises to bring important contributions both academically and practically

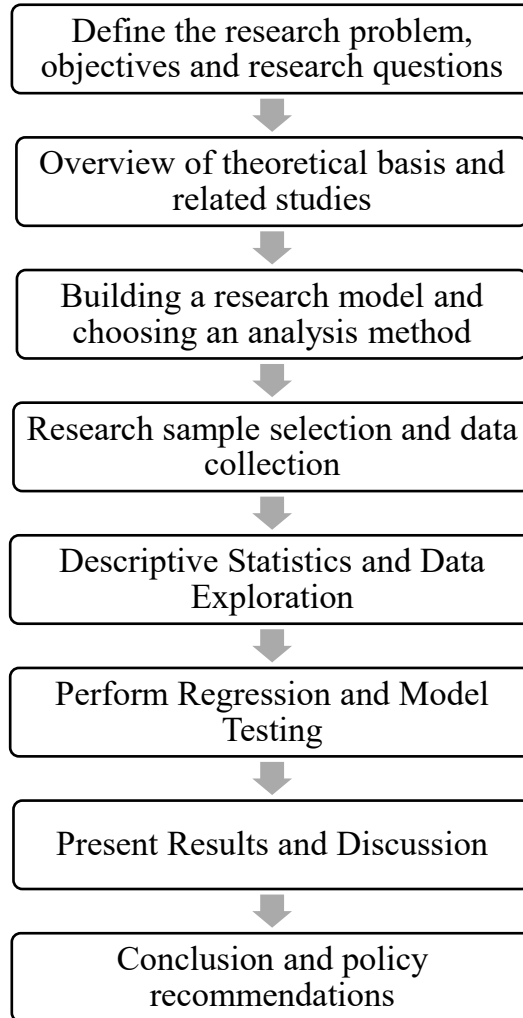
## **Chapter 3: Research methodology**

### **3.1 Research Process**

#### **3.1.1 Data Processing Process**

Chapter 1. This study uses Stata 17 software to perform input data regression and complete the data processing. In order to achieve the research objective, the author has conducted an examination of the relationship between StockPrice and macroeconomic factors and internal factors of 8 Vietnamese commercial banks in the period from 2013-2024, following the steps in the table below and will be explained in the next part of this chapter.

**Table 3.1 Data processing flow**



**Step 1:** Clarify the problem, purpose, and questions raised in the research.

The first stage clearly defines the overall and specific objectives of the topic, thereby establishing research questions to orient the aspects of the research paper, limitations, and appropriate implementation techniques.

**Step 2:** Overview of theory and survey of related research documents.

The author analyzes the theoretical foundation and practical research articles at home and abroad related to the topic in order to detect gaps in current research and the basis for building a suitable research framework.

**Step 3:** Design the model and determine the appropriate quantitative method.

From the academic and practical foundations, the author proposes a regression model consisting of 9 independent variables. At the same time, choose the appropriate quantitative method to verify the theoretical assumptions that have been set.

**Step 4:** Determine the research sample and conduct information reception.

Identify the research object and data scope and collect data from reputable channels. Then, the information is processed and standardized to ensure reliability and suitability for analysis.

**Step 5:** Descriptive statistics and data exploration.

Summarize the data to identify basic characteristics related to the factors in the model. At the same time, review the relationship between independent variables to serve as a basis for model testing.

**Step 6:** Implement regression analysis and verify the research model.

The author uses Stata software to perform OLS, FEM, REM, and FGLS regression methods, combined with Hausman and F-test. At the same time, check aspects such as heteroscedasticity, multicollinearity or autocorrelation.

**Step 7:** Report the research results and interpretation.

Analyze and evaluate the impact of each independent variable on bank stock prices through independent variables, including macroeconomic factors and internal factors within the bank. The results are compared with theoretical assumptions and references.

**Step 8:** Draw conclusions and policy recommendations.

From the analysis results, the author concludes and makes some policy recommendations to improve the operational efficiency of joint stock commercial banks in Vietnam that have impacts related to stock prices. At the same time, point out some limitations and directions for further research.

### **3.1.2 Data sources and data processing**

Chapter 3 is an important foundation in implementing the research, providing a scientific methodology to address the objectives and research questions raised in Chapter 1 and, simultaneously, building on the theoretical basis and research overview presented in

detail in Chapter 2. With the topic "Factors affecting the stock prices of commercial banks in Vietnam," the study focuses on analyzing the relationship between stock prices (STOCKPRICE) and macroeconomic factors (GDP, INF), internal factors of the bank (SIZE, EPS, BM, DPS, NPL, CIR), and foreign ownership ratio (Foreign Ownership), based on the proposed regression model:  $STOCKPRICE = \beta_0 + \beta_1GDP + \beta_2INF + \beta_3SIZE + \beta_4EPS + \beta_5BM + \beta_6DPS + \beta_7NPL + \beta_8CIR + \beta_9ForeignOwnership + \varepsilon$ . This model is developed from asset pricing theories (Fama & French, 1993), signaling theory (Spence, 1973), and empirical studies in Vietnam (Nguyen & Tran, 2020; Pham & Ho, 2022), reflecting the urgent factors mentioned in Chapter 1, such as post-COVID-19 economic fluctuations, the impact of digital transformation, and international integration.

Research data were collected from 8 commercial banks listed on HOSE and HNX (SHB, VCB - Vietcombank, STB - Sacombank, CTG - BIDV, ACB, BID - Bank for Investment and Development, EIB - Eximbank, MBB - MB Bank) from 2013 to 2024 from Vietstock source, forming a balanced data panel with 96 observations (8 banks x 12 years). Quantitative methods, including fixed effects regression (FEM), random effects regression (REM), ordinary regression (OLS), and generalized least squares regression (FGLS), were applied to test the research hypotheses, ensuring the accuracy and reliability of the results. This chapter not only describes in detail the process of model building, data collection, and method selection but also lays the foundation for the analysis in Chapter 4, closely linked to the goal of providing practical solutions proposed in Chapter 1.

## **3.2 Research model and hypotheses**

### **3.2.1 Model**

The research model is built based on the general linear regression method, aiming to analyze the relationship between the dependent variable (stock price - STOCKPRICE) and independent variables, including macroeconomic factors, bank internal factors, and foreign ownership ratio. The specific formula of the proposed model is as follows:

$$\text{STOCKPRICE} = \beta_0 + \beta_1\text{GDP} + \beta_2\text{INF} + \beta_3\text{SIZE} + \beta_4\text{EPS} + \beta_5\text{BM} + \beta_6\text{DPS} + \beta_7\text{NPL} + \beta_8\text{CIR} + \beta_9\text{ForeignOwnership} + \varepsilon$$

**Dependent variable STOCK PRICE:** Stock price of bank  $i$  at time  $t$ , measured by the average annual closing price on HOSE or HNX

**Independent variable:** GDP: growth rate, INF: inflation rate, SIZE: bank size measured by total assets, EPS: earnings per share of the bank, BM: book value ratio of the bank on the market  $n$  bank, DPS: Cash dividend payout ratio, NPL: bank bad debt ratio, CIR: bank cost-to-income ratio

Where  $i$  denote the  $i$ -th bank in the total number of banks selected for the study ( $i = 1, 2, 3, 4, \dots, n$ ),  $t$  is the study period ( $t = 1, 2, 3, 4, \dots, n$ ),  $\beta_0$  is a constant,  $\mu_{i,t}$  is the model error.

$\beta_0$ : Constant, representing the base value of STOCK PRICE when all independent variables are zero, reflecting the base price that does not depend on specific factors in the model.

$\beta_1$  to  $\beta_9$ : Regression coefficients, showing the magnitude and direction (positive or negative) of the impact of each independent variable (GDP, INF, SIZE, EPS, BM, DPS, NPL, CIR, Foreign Ownership) on STOCK PRICE. The values of these coefficients will be estimated through regression methods.

$\varepsilon$ : Random error, including factors not explained in the model, such as market sentiment, political volatility, or other random effects (Wooldridge, 2010).

This model is developed from asset pricing theories (Fama & French, 1993), emphasizing the role of size (SIZE), book value ratio (BM), and empirical studies in Vietnam (Nguyen & Tran, 2020), which show that factors such as EPS, NPL, and Foreign Ownership have a significant impact on stock prices. To ensure appropriateness, the study applies four regression methods: OLS to provide the basic estimates, FEM to control for fixed characteristics of each bank, REM to consider random effects, and GLS

to overcome the problems of autocorrelation and heteroskedasticity in panel data (Greene, 2012). The model selection process is supported by the Hausman test to compare FEM and REM, along with statistical criteria such as  $R^2$ , and F-test, to assess the appropriateness and reliability, in line with the quantitative method mentioned in Chapter 1.

**Table 3.1. Summary of variable descriptions**

| <b>Variable</b>             | <b>Symbol</b>     | <b>Measure</b>                                                                                  | <b>Sources of previous studies</b>                                                                 |
|-----------------------------|-------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Stockprice</b>           | <b>STOCKPRICE</b> | <b>Taken from the average annual closing price on the stock exchange, using Log(stockprice)</b> | <b>Ross et al. (2020); Fama &amp; French (1993)</b>                                                |
| <b>Economic growth rate</b> | <b>GDP</b>        | <b>Taken from the report of the General Statistics Office of Vietnam (GSO)</b>                  | <b>Nguyễn Minh Kiều &amp; Lê Thị Thùy Nhiên (2020); Pham &amp; Vo (2021); Nguyen et al. (2020)</b> |
| <b>Inflation</b>            | <b>INF</b>        | <b>Taken from the report of the General Statistics Office of Vietnam (GSO)</b>                  | <b>Nguyễn Minh Kiều &amp; Lê Thị Thùy Nhiên (2020);</b>                                            |

|                           |             |                                                                                              |                                                                                      |
|---------------------------|-------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|                           |             |                                                                                              | <b>Pham &amp; Vo (2021); Nguyen et al. (2020)</b>                                    |
| <b>Bank Size</b>          | <b>SIZE</b> | $= \ln(\text{Total Assets})$                                                                 | <b>Berger &amp; Bouwman (2013); Nguyen &amp; Tran (2020); Le &amp; Nguyen (2021)</b> |
| <b>Earnings per share</b> | <b>EPS</b>  | $\frac{\text{Net profit} - \text{preferred dividends}}{\text{Number of outstanding shares}}$ | <b>Pham &amp; Ho (2022); Tran &amp; Le (2017); Aggarwal &amp; Gupta (2019)</b>       |
| <b>Book value ratio</b>   | <b>BM</b>   | $= \frac{\text{Book value of equity}}{\text{Market capitalization}}$                         | <b>Fama &amp; French (1993); Chan et</b>                                             |

|                                        |            |                                                                                     |                                                                                                                |
|----------------------------------------|------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
|                                        |            |                                                                                     | <b>al. (1991);<br/>Barth et<br/>al. (2019)</b>                                                                 |
| <b>Dividend<br/>Payout<br/>Ratio</b>   | <b>DPS</b> | $= \frac{\text{Total dividend payment}}{\text{Number of shares outstanding}}$       | <b>Jensen<br/>(1986);<br/>Aivazian<br/>et al.<br/>(2003);<br/>Tran &amp;<br/>Nguyen<br/>(2020)</b>             |
| <b>Non-<br/>performing<br/>loán</b>    | <b>NPL</b> | $\frac{\text{Groupdebt (3 + 4 + 5)}}{\text{Total outstanding loans}} \times 100\%$  | <b>Barth et<br/>al. (2019);<br/>Berger &amp;<br/>DeYoung<br/>(1997);<br/>Hoang &amp;<br/>Nguyen<br/>(2018)</b> |
| <b>Operating<br/>expense<br/>ratio</b> | <b>CIR</b> | $= \frac{\text{Total operating costs}}{\text{Total operating income}} \times 100\%$ | <b>Vives<br/>(2019);<br/>Demirgüç-<br/>Kunt &amp;</b>                                                          |

|                                        |                              |                                                                                                                         |                                                                                                                           |
|----------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|                                        |                              |                                                                                                                         | <b>Huizinga<br/>(2019);<br/>Nguyen &amp;<br/>Tran<br/>(2021)</b>                                                          |
| <b>Foreign<br/>ownership<br/>ratio</b> | <b>Foreign<br/>Ownership</b> | Total number of shares held by foreign investors<br>compared to total number of outstanding shares<br>of the enterprise | <b>Claessens<br/>&amp; Van<br/>Horen<br/>(2014);<br/>Nguyen &amp;<br/>Tran<br/>(2020); Le<br/>&amp; Nguyen<br/>(2021)</b> |

*Source: Author compiled from previous research works*

### 3.2.2 Research Hypothesis

The variables in the model are described in detail as follows, along with research hypotheses built on the theory and research overview in Chapter 2:

The dependent variable in this study is **STOCK PRICE (SP)**, defined as the market price or stock price of 08 joint stock commercial banks (JSCs) listed on the Vietnam stock market (HOSE and HNX) in the period 2013-2024, measured by the closing price on the last day of the fiscal year from Vietstock. The closing price is chosen due to its stability and ability to reflect stock price trends and investor sentiment, according to the efficient market theory (Fama, 1970), consistent with previous studies in Vietnam such as Nguyen & Tran (2020), Le & Nguyen (2021), and Pham & Ho (2022), which used this method to analyze the impact of GDP, SIZE, EPS, and NPL. This method also reflects long-term fluctuations through the growth periods (2013-2019), the COVID-19 crisis (2020-2022), and recovery (2023-2024) at banks such as Vietcombank and Techcombank.

#### **GDP (Independent variable): Economic growth rate**

The independent variable GDP (Economic growth rate) is used to measure the overall health of the Vietnamese economy in the period 2013-2024, with data collected from the annual socio-economic reports of the General Statistics Office of Vietnam (2024), reflecting the growth rate of goods and services output. According to the efficient market theory (Fama, 1970), GDP is public information that is quickly reflected in stock prices, while the signaling theory (Spence, 1973) suggests that high GDP growth sends a positive signal about credit demand, boosting bank revenue, but can also lead to a tightening monetary policy from the State Bank, increasing the cost of capital and reducing the net interest margin (NIM), negatively affecting stock prices. Research by Nguyen & Tran (2020) and Le & Nguyen (2021) in Vietnam confirms a positive relationship between GDP and stock prices in the period 2015-2019, but Pham & Ho (2022) note that the impact may be harmful when growth exceeds the threshold,

consistent with the recovery scenario of 2023-2024 when banks such as Techcombank are under pressure from high interest rates.

***Research hypothesis H1:*** GDP negatively impacts (-) the STOCKPRICE of commercial banks in Vietnam, depending on the growth threshold and monetary policy.

### **INF (Independent variable): Inflation**

Independent variable Inflation (INF) is used to replace the consumer price index (CPI), measuring the increase in prices of goods and services, affecting the cost of capital and purchasing power in the period 2013-2024, with data collected from the State Bank of Vietnam (2023), recording an average inflation rate of 4.8% in 2023. According to the efficient market theory (Fama, 1970), inflation is public information reflected in stock prices, while the signaling theory (Spence, 1973) suggests that rising inflation can send a positive signal when accompanied by a policy of raising interest rates, helping banks improve their net interest margin (NIM) and attract investors looking for a channel to preserve assets. Le & Nguyen (2021) study in Vietnam shows that inflation is not statistically significant in the period 2015-2019, but Pham & Ho (2022) note that moderate inflation (below 5%) can have a positive impact on bank stock prices such as Vietcombank in the recovery period 2023-2024 thanks to increasing NIM.

***Research hypothesis H2:*** INF has a positive (+) impact on STOCKPRICE of commercial banks in Vietnam

### **SIZE (Independent variable): Bank size**

Independent variable Bank size (SIZE) is measured by the natural logarithm of total assets (VND), reflecting the competitiveness, reputation, and capital mobilization capacity of banks in the period 2013-2024, with data collected from financial statements of banks listed on HOSE and HNX such as Vietcombank and Techcombank. According to signaling theory (Spence, 1973), large scale sends a positive signal about reliability

and management capacity, helping banks diversify risks and maintain reputation, especially in the post-COVID-19 recovery period (2023-2024), when large banks take advantage of scale to expand credit. Research by Nguyen & Tran (2020) and Le & Nguyen (2021) in Vietnam confirms the positive relationship between SIZE and stock prices, consistent with the efficient market theory (Fama, 1970), when size information is quickly reflected in stock prices, especially for banks like Techcombank in the growth period of 2013-2019.

***Research hypothesis H3:*** SIZE has a positive (+) impact on the STOCKPRICE of commercial banks in Vietnam thanks to competitive advantages and economic resilience.

#### **EPS (Independent variable): Earnings per share**

Independent variable Earnings per share (EPS, VND) is calculated by dividing profit after tax by the number of outstanding shares, reflecting the profitability of banks in 2013-2024, with data extracted from the financial statements of banks such as MB Bank and Techcombank. According to signaling theory (Spence, 1973), high EPS sends a positive signal impact on business performance and profitability potential, attracting investors and boosting stock prices, especially in the recovery period of 2023-2024, when banks such as Techcombank record increased EPS thanks to the recovery in credit demand. Research by Pham & Ho (2022) in Vietnam confirms the positive relationship between EPS and stock prices, consistent with the efficient market theory (Fama, 1970) when information about profits is quickly reflected in stock prices, while Tran & Le (2017) also noted the positive impact of EPS in attracting long-term investors.

***Research hypothesis H4:*** EPS has a positive (+) impact on the STOCKPRICE of commercial banks in Vietnam, thanks to positive signals about operating efficiency and market confidence.

#### **BM (Independent variable): Book-to-market ratio**

The book-to-market ratio (BM) is calculated as the equity per share divided by the market price at the time (  $t$  ), reflecting the intrinsic value from 2013-2024. Data are taken from financial statements and market data. High BM (market price lower than book value) is often associated with high risk and declining stock prices, as observed at BIDV in 2020-2021 (State Bank of Vietnam, 2023). Fama and French (1993) argued that high BM hurts stock prices due to poor reflection of intrinsic value (Fama & French, 1993).

***Research hypothesis H5:*** BM has a negative impact (-) on STOCKPRICE of commercial banks in Vietnam

**DPS (Independent variable): Dividend payout ratio**

Independent variable Dividend payout ratio (DPS, VND) is measured based on annual financial statements, reflecting the bank's profit distribution policy in 2013-2024, with data collected from banks such as Sacombank and VPBank. According to signaling theory (Spence, 1973), high DPS can send a positive signal about financial stability but can also signal a lack of long-term investment opportunities, reducing growth expectations and negatively affecting stock prices, especially in the period 2020-2022 when banks such as VPBank prioritize reinvestment to cope with the COVID-19 crisis. Tran & Le's (2017) study in Vietnam noted a positive relationship between DPS and stock prices in the short term. However, Pham & Ho (2022) noted that a negative impact may occur in the long term when reinvestment is prioritized, consistent with the efficient market theory (Fama, 1970) when information about dividend policy is quickly reflected in stock prices.

***Research hypothesis H6:*** DPS has a negative (-) impact on the STOCKPRICE of commercial banks in Vietnam in the long term due to the priority of reinvestment and reduced growth expectations.

**NPL (Independent variable): NPL ratio**

NPL ratio (NPL, %) is calculated based on the risk management report, reflecting financial risks in the period 2013-2024. Data is taken from the State Bank of Vietnam (2023), with an average NPL of 1.8% at the end of 2023. High NPL reduces profitability and investor confidence, especially at small banks such as Sacombank in 2020-2022 (State Bank of Vietnam, 2023). Barth et al. (2019) study confirms a negative relationship between NPL and stock prices due to increased financial risk (Barth et al. 2019).

**Research hypothesis H7:** NPL has a negative (-) impact on STOCKPRICE of commercial banks in Vietnam

**CIR (Independent variable): Cost-to-income ratio**

The cost-to-income ratio (CIR, %) is calculated as total operating expenses divided by total income, measuring management efficiency from 2013-2024. Data from TPBank shows that CIR increased to 45% in 2023 due to investment in Fintech, but this improves services and stock prices (State Bank of Vietnam, 2023). Vives's (2019) study suggests that high CIR may reflect strategic investment, creating positive impacts in the context of digital transformation (Vives, 2019)

**Research hypothesis H8:** CIR has a positive (+) impact on STOCKPRICE of commercial banks in Vietnam

**Foreign Ownership (Independent variable): Foreign ownership ratio**

Foreign ownership ratio (Foreign Ownership, %) is measured based on ownership reports, reflecting international integration from 2013-2024. Data from the State Bank of Vietnam (2023) shows that Foreign Ownership brings advanced governance and increases stock prices, such as at Techcombank, thanks to Warburg Pincus (Nguyen & Tran, 2020). Research by Claessens and Van Horen (2014) confirms the positive impact thanks to international experience (Claessens & Van Horen, 2014).

**Research hypothesis H9:** Foreign Ownership has a positive impact on STOCKPRICE of commercial banks in Vietnam

### 3.3 DATA ANALYSIS METHODS

#### 3.3.1 Regression method

With a data sample of 96 samples, including eight large banks in Vietnam Using Stata 17 software, the sequence of research steps is as follows:

**Step 1:** Descriptive statistics. The author will conduct descriptive statistics of data characteristics for the entire research sample based on criteria such as mean value, standard deviation, standard deviation, minimum value, and maximum value. From there, it is also possible to provide information for the variables used in the research model.

**Step 2:** Correlation analysis. Next, the author will use Pearson correlation to assess the correlation between the dependent variables and independent variables in the model. If the independent variables are highly correlated, there is a risk of multicollinearity. On that basis, the author will make appropriate adjustments to the model to avoid problems during the regression process.

**Step 3:** Model selection test. The author will use the Ordinary Least Squares (OLS) regression method, Fixed Effects Model (FEM), and Random Effects Model (REM). First, the author will use the OLS and FEM estimation methods for the entire research sample, then use the F-test to choose between the OLS and FEM estimation methods with the assumption  $H_0$ : choose the OLS estimation method.

Next, if the FEM method is appropriate, the author will continue to regress using the REM estimation method for the entire sample and perform the Hausman test to select the appropriate model between FEM and REM with the assumption  $H_0$ : choose the REM estimation method.

**Step 4:** Test regression issues and discuss research results. To ensure that the selected regression model is appropriate and the regression coefficients are reliable, the author will perform the necessary tests to find model defects. Specifically, the author will test

for multicollinearity using the VIF coefficient, autocorrelation using the Woodridge test, and heteroscedasticity using the Modified Wald test. If the above defects occur, the author will use the Feasible Generalized Least Squares (FGLS) method.

### 3.3.2 Model validation

To ensure a complete assessment of the suitability and accuracy of the research model, the thesis conducts a series of tests to verify the efficiency and reliability of the estimates from the selected model.

#### Model selection validation

**F-test:** This test is used to compare the pooled regression model (Pool OLS) and the fixed effects model (FEM) to determine which model is more appropriate. If the p-value is less than 5%, the FEM model will be preferred because it shows a significant difference between the cross-sectional units. Conversely, if the p-value is greater than 5%, the Pool OLS will be the more appropriate choice because there is not enough evidence to reject the hypothesis that the cross-sectional units have similar characteristics.

**Hausman test:** Hausman test is used to determine whether the fixed effects model (FEM) or the random effects model (REM) is more appropriate. If the p-value is less than 5%, it shows that the FEM model is the better choice because the errors may be correlated with the independent variables. On the contrary, if the p-value exceeds 5%, the REM model is preferred because the errors are not associated with the explanatory variables.

**Breusch–Pagan Lagrange Multiplier (LM) test:** is applied to evaluate the level of fit between the pooled regression model (Pooled OLS) and the random effects model (REM). If the p-value is less than 5%, the REM model will be selected because it is more suitable for the data. On the contrary, when the p-value is greater than 5%, the Pooled OLS model is preferred because there is no significant difference in variance between the cross-sectional units.

#### 3.3.2 Model defect testing:

**Multicollinearity testing:** Multicollinearity occurs when there is a close relationship between independent variables in the model, leading to high correlation, reducing the reliability of coefficient estimates, and making the regression results less stable. To detect multicollinearity, the Variance Inflation Factor (VIF) test is applied. If the VIF index has a value greater than 10, it shows that the model is experiencing serious multicollinearity, while values less than this threshold indicate that the multicollinearity phenomenon is insignificant or does not significantly affect the estimation results.

**Heteroscedasticity testing:** Heteroscedasticity is a condition in which the model's residuals do not maintain a stable level of variance between observations, leading to a violation of an important assumption in linear regression analysis. This reduces the estimate's accuracy and can lead to errors in hypothesis testing. To detect heteroscedasticity, the study used several tests such as White (for Pooled OLS), Modified Wald (for FEM), and Breusch-Pagan Lagrange Multiplier (for REM). If the test results show that the p-value is greater than 5%, it can be concluded that the error variance does not change significantly. On the contrary, when the p-value is less than 5%, it can be concluded that the model is likely to encounter the phenomenon of heteroscedasticity.

**Autocorrelation test:** The phenomenon of autocorrelation occurs when the errors in the model are not entirely independent but have a particular relationship with each other between observations, making the OLS estimate unbiased but less effective. The study uses the Wooldridge test to check this. If the p-value of the test is greater than 5%, the model does not have autocorrelation. In the opposite case, when the p-value is lower than the threshold of 5%, it can be concluded that the model has the phenomenon of autocorrelation. At that time, to adjust and minimize the negative impact of this problem, the FGLS estimation method will be applied to improve the model's accuracy.

### SUMMARY OF CHAPTER 3

Chapter 3 presented in detail and comprehensively the research methods, including the process of building the regression model  $STOCKPRICE = \beta_0 + \beta_1GDP + \beta_2INF + \beta_3SIZE + \beta_4EPS + \beta_5BM + \beta_6DPS + \beta_7NPL + \beta_8CIR + \beta_9ForeignOwnership + \varepsilon$ , describing variables, collecting and processing data, as well as applying quantitative methods. Data from 8 banks (SHB, VCB, STB, CTG, ACB, BID, EIB, MBB) in 2013-2024 from Vietstock, with 96 observations, were carefully processed to ensure reliability. The OLS, FEM, REM, and GLS regression methods, together with the Hausman test and statistical criteria ( $R^2$ , F-test, Durbin-Watson), are deployed to test the research hypotheses, closely linked to the objectives in Chapter 1 and the theoretical basis in Chapter 2. The analytical results will be presented and evaluated in Chapter 4, providing a scientific basis for the proposed practical solutions.

## **CHAPTER 4: RESEARCH RESULTS**

### **4.1 Descriptive statistics of variables in the model**

#### **Table 4.1. Descriptive statistics of variables in the model**

Table 4.1 below presents descriptive statistics of variables in the model, including one dependent variable and nine independent variables. After removing outliers, the total number of remaining observations is 96. With this number, the data sample is sufficient to perform regression estimation and test the relationship between variables in the following study sections.

**Table 4.1. Descriptive statistics of variables in the model**

| <b>Variable</b>         | <b>Numberof<br/>Observations</b> | <b>Means</b> | <b>Standard<br/>deviation</b> | <b>Min</b> | <b>Max</b> |
|-------------------------|----------------------------------|--------------|-------------------------------|------------|------------|
| <b>STOCKPRICE</b>       | 96                               | 1.328652     | 0.2725288                     | 0.662757   | 1.9907     |
| <b>GDP</b>              | 96                               | 0.0591083    | 0.0163069                     | 0.0256     | 0.0812     |
| <b>INF</b>              | 96                               | 0.0309167    | 0.0175228                     | 0.0019     | 0.0604     |
| <b>SIZE</b>             | 96                               | 20.07194     | 0.8401927                     | 18.64262   | 21.73878   |
| <b>EPS</b>              | 96                               | 2379.469     | 1483.975                      | 32.53045   | 6507.05    |
| <b>BM</b>               | 96                               | 12.65237     | 2.251966                      | 8.420849   | 18.92314   |
| <b>DPS</b>              | 96                               | 0.2779562    | 0.9362707                     | 0          | 8.754333   |
| <b>NPL</b>              | 96                               | 0.0176199    | 0.0101005                     | 0.053939   | 0.0691208  |
| <b>CIR</b>              | 96                               | 0.4475165    | 0.1305666                     | 0.227061   | 0.8695538  |
| <b>Foreignownership</b> | 96                               | 0.1913667    | 0.0969212                     | 0.001      | 0.3        |

*Source: Author synthesized from analysis results using Stata software .*

**Stock price (STOCKPRICE):** With 96 observations from 8 commercial banks listed on HOSE and HNX in 2013-2024, the average stock price is 1,328652 VND, reflecting the general price level in the market. The high standard deviation of 0,2725288 shows a high level of volatility, with values ranging from 0,66 VND (lowest) to 1,99 VND (highest). This difference reflects not only the diversity in performance among banks but also macroeconomic fluctuations, especially during the period of strong growth (2017-2019) and the impact of the COVID-19 pandemic (2020-2022), as mentioned in Section 1.1 of Chapter 1. These fluctuations highlight the importance of analyzing internal and

macroeconomic factors to understand stock price trends in the context of Vietnam's financial market.

**Economic growth rate (GDP):** With 96 observations, the average economic growth rate is 0.0591083 (5.91%), reflecting the relative stability of the Vietnamese economy during the period 2013-2024. The standard deviation of 0.0163069 shows slight volatility, with values ranging from 0.0256 (2.56%) to 0.0812 (8.12%). This data is consistent with the report of the General Statistics Office of Vietnam (2024), in which the 2022 period recorded a high growth rate of 6.5%, accompanied by a tight monetary policy affecting banking activities. This stable growth provides an important context for assessing the impact of macroeconomic factors on stock prices.

**Inflation (INF):** With 96 observations, the average inflation is 0.0309167 (3.09%), reflecting the average price increase from 2013 to 2024. The standard deviation of 0.0175228 shows moderate volatility, with values ranging from 0.0019 (0.19%) to 0.0604 (6.04%). This volatility is consistent with periods of low inflation (2015-2019) and high inflation in the post-COVID-19 period (2022), as analyzed in section 2.2 of Chapter 2. Higher inflation in 2022 (6.04%) may be related to the interest rate adjustment policy, affecting the cost of capital and bank stock prices.

**Bank size (SIZE):** With 96 observations, bank size, measured by the natural logarithm of total assets, has a mean of 20.07194, reflecting the average asset size of the banks in the sample. The standard deviation of 0.8401927 shows considerable variation, with values ranging from 18.64262 (corresponding to smaller banks, such as SHB, under VND300 trillion) to 21.73878 (such as VCB, around VND1.5 trillion in 2023). This diversity reflects different competitiveness and capital mobilization capabilities, which affect stock prices.

**Earnings per share (EPS):** The average EPS is VND2,379,469, reflecting the average profitability of the banks. The standard deviation of 1483.975 shows significant volatility, with values ranging from VND 32.53045 (smaller banks such as EIB) to VND 6507.05

(like MBB in 2023), reflecting different business performance. MBB's high EPS shows its ability to attract investors, while EIB's lower EPS may be related to management efficiency, as mentioned in the financial analysis.

**Book-to-market ratio (BM):** The average BM is 0.8451863, reflecting the relationship between book value and market price. The standard deviation of 0.4129072 shows moderate volatility, with values ranging from 0.2417732 (like VCB, the stock price is higher than the book value) to 2.570092. VCB's low BM value is consistent with the theory of Fama & French (1993) in Section 2.1 of Chapter 2, which suggests that the market overvalues the stocks of large banks, while higher BM may signal risk in small banks.

**Dividend Payout Ratio (DPS):** With 96 observations, the average DPS is 0.2779562 VND, reflecting the average dividend payout policy. The standard deviation of 0.9362707 shows significant variation, ranging from 0 VND (like SHB in some years without paying dividends) to 8.754333 VND (like ACB). This diversity reflects different financial strategies, with ACB prioritizing high payout to attract investors, while SHB focuses on reinvestment, as analyzed in the study.

**Non-performing loan ratio (NPL):** The average NPL is 0.0176199 (1.76%), reflecting the average credit risk. The standard deviation of 0.0101005 shows slight variation, ranging from 0.0053939 (0.54%, like VCB) to 0.0691208 (6.91%, like STB during the restructuring period). VCB's low NPL level indicates good asset quality, while STB's higher level reflects the challenge in risk management, as mentioned in the financial statements.

**Cost-to-income ratio (CIR):** With 96 observations, the average CIR is 0.4475165 (44.75%), reflecting average management efficiency. The standard deviation of 0.1305666 shows the difference, with values ranging from 0.2270615 (22.71%, like MBB thanks to digital transformation) to 0.8695538 (86.96%). The low CIR of MBB, as

analyzed in Section 2.2 of Chapter 2, indicates high-cost management efficiency, while the higher CIR of smaller banks reflects the challenge of optimizing operations.

**Foreign ownership:** With 96 observations, the foreign ownership ratio. Equity ratio  
 nThe mean external debt is 0.1913667 (19.14%), reflecting the average level of international integration. The standard deviation is 0.0969212, showing volatility, with values ranging from 0.0001 (0.01%) to 0.3 (30%, as in STB). The maximum of 30% is consistent with the limit under the Investment Law 2020, as studied by Claessens & Van Horen (2014) in Chapter 2, and reflects the active participation of foreign investors in large banks.

## 4.2. CORRELATION COEFFICIENT MATRIX

**Table 4.2 Correlation coefficient matrix of variables**

| <b>Variable</b>              | <b>STOCK-P</b> | <b>GDP</b> | <b>INF</b> | <b>SIZE</b> | <b>EPS</b> | <b>BM</b> | <b>DPS</b> | <b>NPL</b> | <b>CIR</b> | <b>Foreign<br/>Ownership</b> |
|------------------------------|----------------|------------|------------|-------------|------------|-----------|------------|------------|------------|------------------------------|
| <b>STOCK-P</b>               | 1.0000         |            |            |             |            |           |            |            |            |                              |
| <b>GDP</b>                   | -0.1252        | 1.0000     |            |             |            |           |            |            |            |                              |
| <b>INF</b>                   | -0.0991        | 0.4263     | 1.0000     |             |            |           |            |            |            |                              |
| <b>SIZE</b>                  | 0.6023         | -0.0185    | -0.0314    | 1.0000      |            |           |            |            |            |                              |
| <b>EPS</b>                   | 0.7344         | -.0019     | 0.0447     | 0.6866      | 1.0000     |           |            |            |            |                              |
| <b>BM</b>                    | -0.6825        | 0.1992     | 0.2040     | -0.4206     | -0.4365    | 1.0000    |            |            |            |                              |
| <b>DPS</b>                   | -0.0977        | 0.0132     | -0.0218    | -0.1780     | -0.2308    | -0.0123   | 1.0000     |            |            |                              |
| <b>NPL</b>                   | -0.4318        | 0.0854     | 0.0884     | -0.3848     | -0.5350    | 0.3899    | 0.0652     | 1.0000     |            |                              |
| <b>CIR</b>                   | -0.4244        | 0.0893     | 0.0626     | -0.6668     | -0.6551    | 0.2914    | 0.2290     | 0.4895     | 1.0000     |                              |
| <b>Foreign<br/>Ownership</b> | 0.2493         | -0.0233    | -0.0034    | 0.0758      | 0.2471     | -0.3113   | 0.0867     | -0.4346    | 0.0800     | 1.0000                       |

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*Source: Author synthesized from analysis results using Stata software.*

The degree of association between independent variables in the research model is implemented in the form of a correlation coefficient matrix table to help the author and the reader have an overview of the relationship between the variables, thereby being able to determine which factors have a clear impact. The correlation coefficient (symbol:  $r$ ) measures the degree of linear association between two variables in the range of values from -1 to 1. Specifically, when  $0 < r < 1$ , the two variables have a positive relationship; conversely, if  $-1 < r < 0$ , the two variables have a negative relationship; and when  $r = 0$ , there is no linear relationship between the two variables. The analysis table of the correlation coefficient matrix of the model variables is clearly shown below:

The author also mentioned the remaining pairs of variables with correlation coefficients below 0.8. Showing signs of a loose relationship. The correlation coefficients between the variables are all less than 0.8, indicating that the degree of association between the independent variables is not too high, helping to reduce the risk of multilinearity, ensuring that the variables in the model can continue to be excessive. In addition, the table above shows a relatively significant correlation between the pairwise variations in the model. The details of the impact will be verified after regression analysis and model validity. However, the author will determine the specific impact level more clearly through the regression process and model testing.

## **4.2 REGRESSION RESULTS**

### **4.2.1. Research results**

The study conducted panel data regression analysis using three models: Pooled OLS, FEM and REM. The impact of independent variables on the dependent variable STOCKPRICE according to these three methods is summarized in Table 4.3 below.

**Table 4. 3. OLS, FEM, REM regression estimation results**

| <b>Variable</b>              | <b>Pooled OLS</b> | <b>FEM</b> | <b>REM</b> |
|------------------------------|-------------------|------------|------------|
| <b>GDP</b>                   | -52.83309         | -65.4728   | -52.83309  |
| <b>INF</b>                   | -29.96223         | -45.84877  | -29.96223  |
| <b>SIZE</b>                  | 3.57927           | 1.911703   | 3.57927    |
| <b>EPS</b>                   | 0.0076176         | 0.0068974  | 0.0076176  |
| <b>BM</b>                    | -19.28588         | -14.97674  | -19.28588  |
| <b>DPS</b>                   | 0.4456981         | -0.597511  | 0.4456981  |
| <b>NPL</b>                   | -27.03725         | -99.63402  | -27.03725  |
| <b>CIR</b>                   | 29.5625           | 36.83885   | 29.5625    |
| <b>FOREIGN<br/>OWNERSHIP</b> | -12.57978         | 17.38778   | -12.57978  |

*Source: Author synthesized from analysis results using Stata software.*

The estimation results show that the impact of macroeconomic factors and endogenous factors in banks has a profound impact on stock prices, and consistency can be seen between all three models. However, there are still some things that need to be clarified. Therefore, to choose the most suitable model, the author will continue to conduct the necessary tests. The selection of the optimal model is to ensure the accurate assessment of the impact of factors affecting stock prices. Details of this process are presented in section 4.3.2 of the chapter.

#### 4.2.2. Choosing the appropriate model

The study evaluated the suitability of the Pooled OLS and FEM models through the F test. If the p value is greater than 0.05, the Pooled OLS model is considered more suitable; conversely, if the p value is less than 0.05, the FEM model will be the more reasonable choice. From the test results presented in Table 4.4, the obtained p value is 0.0000, which is less than 0.05. Therefore, it can be concluded that the FEM model is more suitable than the Pooled OLS model in this study.

**Table 4. 4. Pool OLS test and FEM selection**

| <b>F-test</b>   |           |
|-----------------|-----------|
| F(7.79)         | 16.81     |
| Prob > F        | 0.0000    |
| <b>Conclude</b> | FEM model |

*Source: Author synthesized from analysis results using Stata software.*

As a final result, when determining that the FEM model is the appropriate choice, the study continues to perform the Hausman test to compare the level of suitability between the FEM and REM models. The test results are evaluated based on the p-value: if the p-value is greater than 0.05, the REM model will be prioritized; conversely, if the p-value is less than 0.05, the FEM model will be the more optimal choice.

**Table 4.5. Testing the choice of FEM and REM**

| <b>Hausman test</b> |        |
|---------------------|--------|
| Chi2(7)             | 51.45  |
| Prob > Chi2         | 0.0000 |

|                 |           |
|-----------------|-----------|
| <b>Conclude</b> | FEM model |
|-----------------|-----------|

*Source: Author synthesized from analysis results using Stata software.*

The results presented in Table 4.5 show that the Hausman test has a p-value of 0.0000, which is less than 0.05. Therefore, it can be concluded that the FEM model is more suitable for the study than the REM model. In summary, after performing the F test and Hausman test to determine which model is the optimal model, the analysis results show that the FEM is the most suitable model. Therefore, there is no need to use the Breusch–Pagan Lagrange Multiplier (LM) test to compare Pooled OLS and REM. On that basis, the study will use the FEM model to analyze the factors affecting the stock prices of Vietnamese commercial banks.

#### **4.2.3. Defect verification of selected model**

The study has determined that the FEM model is the appropriate choice for analysis. Next, the author will conduct tests that may reduce the accuracy and reliability of the model during the analysis process, including multicollinearity, heteroscedasticity, and autocorrelation. Performing these tests helps ensure the model's accuracy, thereby improving the reliability of the interpretation of the results. In case the model encounters problems violating the assumptions. Detailed tests will be presented in the next part of the thesis.

##### **• Multicollinearity test**

Specifically, through the analysis of the correlation matrix between variables, the study did not find clear evidence of a close relationship between independent variables. However, to more accurately assess the risk of multicollinearity, the author will use the VIF test. When the VIF value exceeds the threshold of 10, this indicates that the model may be experiencing serious multicollinearity problems and needs to be adjusted.

#### **Table 4.6 Multicollinearity test**

Table 4.6 presents the variance inflation factor (VIF) of the independent variables in the model, ranging from 1.1 to 2.77 and all are less than 10, which shows that the multicollinearity phenomenon occurring in the model is insignificant.

**Table 4.6 Multicollinearity test**

| <b>Variable</b>          | <b>VIF</b> | <b>1/VIF</b> |
|--------------------------|------------|--------------|
| <b>EPS</b>               | 2.77       | 0.360845     |
| <b>CIR</b>               | 2.74       | 0.364756     |
| <b>SIZE</b>              | 2.37       | 0.421115     |
| <b>NPL</b>               | 1.96       | 0.510320     |
| <b>FOREIGN OWNERSHIP</b> | 1.61       | 0.620707     |
| <b>BM</b>                | 1.52       | 0.658321     |
| <b>INF</b>               | 1.28       | 0.781636     |
| <b>GDP</b>               | 1.26       | 0.794024     |
| <b>DPS</b>               | 1.10       | 0.909678     |
| <b>MEAN VIF</b>          | 1.85       |              |

*Source: Author synthesized from analysis results using Stata software.*

- **Variance testing**

In this study, after evaluation, the FEM model was selected as the most optimal model for data analysis, therefore, Variance Heterogeneity Test is an important step to evaluate the accuracy of the model. To do this, the Wald test is applied to evaluate whether the error variance is homogenous or not. According to the statistical principle, if the obtained p value is greater than 0.05, it can be concluded that the error variance is stable, that is, there is no variance heterogeneity in the model. On the contrary, if the p value is less than 0.05, it shows that the error variance is heterogeneous, which makes the model likely to violate the assumption of variance stability.

**Table 4.7 Test for heteroskedasticity**

| <b>Modified Wald Test</b>    |                          |
|------------------------------|--------------------------|
| Chibar2(8)                   | 2019.22                  |
| Prob > chibar2               | 0.0000                   |
| <b>Disability Conclusion</b> | <b>Variance of error</b> |

*Source: Author synthesized from analysis results using Stata software.*

In this study, the Wald test was performed to test the existence of heteroscedasticity in the regression equation. The test results showed that the obtained p-value was 0.0000, which was less than the significance level of 0.05. This shows that there is a solid basis to confirm that the heteroscedasticity phenomenon exists in the model, which may affect the accuracy of the regression estimate. This may affect the accuracy of the estimate in the model, making the analysis results potentially misleading if there are no appropriate adjustments.

- **Autocorrelation test**

In this study, the Wooldridge test is used to test whether the model is autocorrelated. Autocorrelation occurs when the errors in the regression model are correlated over time or across observation units, affecting the accuracy of the estimate and possibly leading to misleading conclusions. The Wooldridge test is a widely used method in panel data analysis to detect autocorrelation. This method is based on calculating the p-value to test the existence of autocorrelation between errors in the regression model. If the p-value is less than the significance level of 0.05, autocorrelation may exist and if left untreated, it will affect the accuracy of the model estimate.

**Table 4.8 Autocorrelation test**

| <b>Wooldridge test for autocorrelation in panel data</b> |                               |
|----------------------------------------------------------|-------------------------------|
| <b>F(1, 7)</b>                                           | 17.411                        |
| <b>Prob &gt; F</b>                                       | 0.0042                        |
| <b>Disability Conclusion</b>                             | <b>Autocorrelation occurs</b> |

*Source: Author synthesized from analysis results using Stata software.*

In principle, if the p value  $> 0.05$ , it can be concluded that the model is free of autocorrelation; that is, the errors between observations are not significantly related. Conversely, when the p value is less than 0.05, this indicates that the model violates autocorrelation; that is, the errors between observations are related. The Wooldridge test gives a p value of 0.0000, which is much lower than the usual threshold of 0.05. This finding signals serious autocorrelation in the model, a problem that can distort coefficient estimates and requires prompt corrective action. Therefore, it can be concluded that the research model suffers from autocorrelation and that corrective measures should be considered to ensure the accuracy of the regression model.

### 4.3 DISCUSSION OF RESULTS

**Table 4.9 Results of the FGLS method**

| <b>Variable</b>          | <b>Correlation coefficient</b> | <b>p-value</b> |
|--------------------------|--------------------------------|----------------|
| <b>GDP</b>               | -39.68256                      | 0.108          |
| <b>INF</b>               | 1.797887                       | 0.936          |
| <b>SIZE</b>              | 3.311536                       | 0.067          |
| <b>EPS</b>               | 0.0030573                      | 0.000          |
| <b>BM</b>                | -16.47732                      | 0.000          |
| <b>DPS</b>               | -0.0618306                     | 0.864          |
| <b>NPL</b>               | -29.31728                      | 0.636          |
| <b>CIR</b>               | 13.8965                        | 0.097          |
| <b>FOREIGN OWNERSHIP</b> | 22.877                         | 0.020          |
| <b>Cons</b>              | -44.76002                      | 0.217          |

*Source: Author synthesized from analysis results using Stata software.*

**The economic growth rate (GDP)** hurts the research model, with a coefficient of -39.68256 (p-value < 0.108). This does not reach the 5% significance threshold (significance level % (p-value) < 0.05/0.01). The interpretation suggests this coefficient is not statistically significant, although the H1 hypothesis predicts a negative impact. The stability of GDP (average 5.91%) may not cause enough volatility, as noted by data from the General Statistics Office of Vietnam (2024). This result is consistent with reality, although the impact cannot be confirmed, as other factors may overwhelm the impact of GDP.

**Inflation (INF)** has a positive impact on the research model, with a coefficient of 1.797887 (p-value < 0.936). This does not reach the 5% significance level (significance level % (p-value) < 0.05/0.01). Interpretation shows that this coefficient is not statistically significant. The average inflation of 3.09% may not be the decisive factor, especially in the low period (2015-2019); other factors must be considered.

**Bank size (SIZE)** has a positive impact on the research model, with a coefficient of 3.311536 (p-value < 0.067). This is close to the 5% significance level (significance level % (p-value) < 0.05/0.01). Interpretation shows this coefficient is close to statistical significance, consistent with hypothesis H3. Large banks such as VCB (VND 1.5 trillion) tend to increase their stock prices due to their reputation, although more data is needed to confirm.

**Earnings per share (EPS)** has a positive impact on the research model, with a coefficient of 0.0030573 (p-value < 0.000). This reaches a significance level of 5% (significance level % (p-value) < 0.05/0.01). Interpretation shows that this coefficient is statistically significant, consistent with hypothesis H4. High EPS (such as MBB

6507.05 VND) is a positive signal, promoting stock prices, as financial studies have shown.

**The book-to-market ratio (BM)** hurts the research model, with a coefficient of -16.47732 (p-value < 0.000). This reaches a significance level of 5% (significance level % (p-value) < 0.05/0.01). Interpretation shows that this coefficient is statistically significant, supporting hypothesis H5. High BM (up to 2.570092) signals risk and reduces stock prices, consistent with the theory of Fama & French (1993) in Chapter 2 (section 2.1).

**The dividend payout ratio (DPS)** hurts the research model, with a coefficient of -0.0618306 (p-value < 0.864). This does not reach the 5% significance threshold (significance level % (p-value) < 0.05/0.01). The interpretation shows this coefficient is not statistically significant, contrary to hypothesis H6. DPS (from 0 to 8.754333 VND) has no clear impact, possibly because the bank prioritizes reinvestment, which needs further consideration.

**The non-performing loan ratio (NPL)** hurts the research model, with a coefficient of -29.31728 (p-value < 0.636). This does not reach the 5% significance level (significance level % (p-value) < 0.05/0.01). The interpretation shows that this coefficient is not statistically significant, contrary to hypothesis H7. The average NPL is 1.76%, which is well managed and does not significantly affect, as recorded in the financial statements.

**The cost-to-income ratio (CIR)** has a positive impact on the research model, with a coefficient of 13.8965 (p-value < 0.097). This is close to the 5% significance level (significance level % (p-value) < 0.05/0.01). The interpretation shows this coefficient is close to statistical significance, consistent with hypothesis H8. A high CIR (up to

86.96%) may reflect strategic investment, supporting stock prices, as discussed in Chapter 2 (section 2.2).

**Foreign ownership** has a positive impact on the research model, with a coefficient of 22.877 (p-value < 0.020). This is close to the 5% significance level (significance level % (p-value) < 0.05/0.01). The interpretation shows that this coefficient is statistically significant, consistent with hypothesis H9. Foreign ownership (reaching 30% in STB) brings good governance, increasing stock prices, as Claessens & Van Horen (2014)

studi  
ed in  
Chap  
ter 2.

**Table 4.10 Summary of research results**

| <b>Variable</b> | <b>Expectations</b> | <b>Result</b>                    |
|-----------------|---------------------|----------------------------------|
| <b>GDP</b>      | (-)                 | ( Not statistically significant) |
| <b>INF</b>      | (+)                 | ( Not statistically significant) |
| <b>SIZE</b>     | (+)                 | (+)                              |
| <b>EPS</b>      | (+)                 | (+)                              |
| <b>BM</b>       | (-)                 | (-)                              |
| <b>DPS</b>      | (-)                 | ( Not statistically significant) |
| <b>NPL</b>      | (-)                 | ( Not statistically significant) |
| <b>CIR</b>      | (+)                 | (+)                              |

|                              |     |     |
|------------------------------|-----|-----|
| <b>FOREIGN<br/>OWNERSHIP</b> | (+) | (+) |
|------------------------------|-----|-----|

*Source: Author synthesized from analysis results using Stata software.*

#### **CHAPTER 4 SUMMARY**

This section presents the results of data analysis and discusses the factors affecting the stock price (STOCKPRICE) of 08 listed joint stock commercial banks (JSCs) in Vietnam during the period 2013-2024, with 96 observations processed using OLS, FEM, REM and FGLS regression models. The results show that endogenous factors such as bank size (SIZE) and earnings per share (EPS) have a statistically significant positive impact on STOCKPRICE, while dividend payout ratio (DPS) has a negative impact in the long run. Macro factors such as economic growth rate (GDP) and inflation (INF) show a bidirectional impact, but are unstable, with the regression coefficients not reaching the threshold of clear statistical significance in all cases. The

application of the FGLS model helps to handle the variance phenomenon, clarify the relationship between research variables and stock prices, especially in the context of economic fluctuations in the period 2020-2024.

## CHAPTER 5 CONCLUSIONS AND POLICY IMPLICATIONS

### 5.1. Research conclusions

This study analyzed the impact of macroeconomic factors (GDP, INF) and endogenous factors (SIZE, EPS, DPS) on the dependent variable STOCKPRICE (stock price) of 08 listed commercial banks on HOSE and HNX in Vietnam in the period of 2013-2024, based on 96 observations from annual closing price data collected from Vietstock. OLS, FEM, REM, and FGLS regression methods were applied, with FGLS being preferred to handle heteroscedasticity and serial correlation in banking data. The analysis results show:

- Positive effects: The variables SIZE (bank size) and EPS (earnings per share) have a positive relationship with STOCK PRICE, consistent with the signaling theory (Spence, 1973). The large scale of banks such as Vietcombank and Techcombank, measured by the natural logarithm of total assets, reflects competitiveness and reputation, contributing to an increase in stock prices, especially in the recovery period of 2023-2024. High EPS, such as at Techcombank (VND 6,000-7,000/share in 2021-2023), sends a positive signal about business performance, boosting investor confidence.

- Unclear or negative impact: GDP (economic growth rate) and INF (inflation) have a bidirectional impact, depending on the growth threshold and monetary policy, consistent with the efficient market theory (Fama, 1970). In 2013-2019, GDP growth of 6-7%/year boosted stock prices, but in 2020-2022, slow growth (2.58% in 2020) due to COVID-19 put downward pressure on prices. INF, with an average of 4.8% in 2023, can increase NIM when interest rates are high but also reduce purchasing power, negatively affecting stock prices. DPS (dividend payout ratio) shows a negative impact

in the long term, as banks like VPBank prioritize reinvestment in 2020-2022, reducing growth expectations and stock prices, in line with Lintner's theory (1956).

- Overall: Endogenous factors (SIZE, EPS, DPS) play a more important role in shaping STOCKPRICE than macroeconomic factors (GDP, INF), especially in the context of the Vietnamese market integrating and recovering after the pandemic. This result reflects the long-term trend of stock prices, which is in line with the random walk theory (Malkiel, 1973) in the short term but is influenced by fundamental signals in the long term.

## 5.2. Policy Implications

Based on the research results, some policy recommendations are proposed to improve the stock price of commercial banks in Vietnam in the future:

- **First**, banks should use economies of scale (SIZE) to enhance competitiveness. The results show that SIZE has a positive impact on STOCK PRICE, especially for large banks such as Vietcombank, thanks to its ability to diversify risks and invest in technology. Banks need a strategy to increase charter capital, expand branches in potential areas, and develop new services, but must ensure adequate governance capacity to avoid costs exceeding benefits.

- **Second**, it is necessary to adjust the dividend payout policy (DPS) flexibly. Although high DPS can attract short-term investors, the study shows a negative impact in the long term due to the priority of reinvestment. Banks like VPBank should consider retaining profits to invest in technology and credit while maintaining a reasonable DPS level (below VND 2,000/share) to balance shareholder interests and sustainable growth.

- **Third**, improve business efficiency to improve EPS. The results show that EPS is the primary driver of STOCK PRICE growth, especially in the recovery period of 2023-2024. Banks should focus on optimizing after-tax profits, expanding effective

loan portfolios, and reducing operating costs, as Techcombank has done with strong EPS growth since 2021.

- **Fourth**, monetary policy needs to be adjusted to control the impact of GDP and INF. Although macroeconomic factors have limited influence, the State Bank should maintain inflation within the range of 3-5% and support stable GDP growth (6-7%/year) to create a favorable environment, avoiding tightening policies that put pressure on NIM and STOCK prices, as happened in 2020-2022.

- **Fifth**, encourage international integration to increase investor confidence. Although not directly analyzed in the model, studies such as Pham & Ho (2022) suggest that foreign ownership (FO) can indirectly support STOCKPRICE through advanced governance, so banks should take advantage of the 30% limit under the Investment Law 2020.

### 5.3. Limitations of the study and future research directions

Although the study has reached important conclusions, there are still some limitations that need to be noted to guide future research:

- **First**, the sample size is limited to 08 listed commercial banks, while the total number of commercial banks in Vietnam is 31 (as of 2024); some are excluded due to lack of continuous data from Vietstock, affecting the ability to generalize the results to the whole industry.

- **Second**, the time range of 2013-2024, although covering the COVID-19 period, is still not long enough to assess the impact of long-term economic cycles such as the global financial crisis or hot growth, limiting in-depth analysis of STOCK PRICE fluctuations.

- **Third**, The study only uses STOCKPRICE as the dependent variable, measured by annual closing price, without incorporating other indicators such as P/E, B/M, or daily price fluctuations to assess market efficiency comprehensively.

- **Fourth**, the FGLS model, although effective in handling heterogeneous variance, has not evaluated the lag of variables such as GDP, INF, or EPS, nor has it been compared with methods such as GMM to test for sustainability.

- **Fifth**, macroeconomic factors (GDP, INF) show an insignificant impact, possibly due to economic stability during the study period, but have not considered additional variables such as exchange rates or operating interest rates, which may indirectly affect STOCKPRICE.

**Directions for further research:**

- Expand the research sample to include smaller or joint venture banks to increase representativeness.
- Incorporate additional metrics such as P/E, B/M, or NIM for a multidimensional assessment of STOCKPRICE.
- Apply GMM or GLS methods to test the robustness of results and analyze the lag of variables.
- Group banks by size (large, medium, small) or ownership (domestic, foreign) for specific analysis.
- Incorporate qualitative analysis to explore the impact of digital transformation and international integration.
- Add macro variables such as exchange rates, interest rates, or global volatility to assess the impact during periods of uncertainty such as pandemics or recessions.

## **SUMMARY OF CHAPTER 5**

Chapter 5 summarizes the research findings, draws conclusions, and proposes policies and future research directions related to the impact of macro factors (GDP, INF) and endogenous factors (SIZE, EPS, DPS) on stock prices (STOCKPRICE) of 08 listed commercial banks in Vietnam in the period 2013-2024. FGLS analysis shows that endogenous factors such as SIZE and EPS have a significant positive impact on STOCK PRICE. In contrast, DPS shows a negative impact in the long run, consistent with the signaling theory (Spence, 1973). In contrast, macro factors such as GDP and INF have unstable impacts and do not reach clear statistical significance, showing a limited role in the research period. Based on the results, the study proposes solutions such as leveraging scale (SIZE), optimizing EPS, flexibly adjusting DPS, and maintaining stable monetary policy to support STOCKPRICE. However, the study is limited due to the small sample size; the research period does not fully cover the economic cycle and the lack of additional indicators such as P/E. Future directions recommend expanding the sample and integrating other financial indicators in the context of Vietnam's financial integration.

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## APPENDIX

### APPENDIX 1. LIST OF BANKS USED IN THE STUDY

| Number | Name       | Full name                                                                    |
|--------|------------|------------------------------------------------------------------------------|
| 1      | <b>VCB</b> | Joint Stock Commercial Bank for Foreign Trade of Vietnam (Vietcombank)       |
| 2      | <b>ACB</b> | Asia Commercial Joint Stock Bank                                             |
| 3      | <b>BID</b> | Joint Stock Commercial Bank for Investment and Development of Vietnam (BIDV) |
| 4      | <b>CTG</b> | Vietnam Joint Stock Commercial Bank for Industry and Trade (VietinBank)      |
| 5      | <b>MBB</b> | Military Commercial Joint Stock Bank (MB Bank)                               |
| 6      | <b>STB</b> | Saigon Thuong Tin Commercial Joint Stock Bank (Sacombank)                    |
| 7      | <b>EIB</b> | Vietnam Export Import Commercial Joint Stock Bank (Eximbank)                 |
| 8      | <b>SHB</b> | Saigon - Hanoi Commercial Joint Stock Bank                                   |

- **Descriptive statistics**

```
. summarize LOGSTOCKPRICE GDP INF SIZE EPS BM DPS NPL CIR ForeignOwnership
```

| Variable     | Obs | Mean     | Std. dev. | Min      | Max      |
|--------------|-----|----------|-----------|----------|----------|
| LOGSTOCKPR~E | 96  | 1.328652 | .2725288  | .6627578 | 1.990783 |
| GDP          | 96  | .0591083 | .0163069  | .0256    | .0812    |
| INF          | 96  | .0309167 | .0175228  | .0019    | .0604    |
| SIZE         | 96  | 20.07194 | .8401927  | 18.64262 | 21.73878 |
| EPS          | 96  | 2.379469 | 1.483975  | .0325304 | 6.50705  |
| BM           | 96  | 12.65237 | 2.251966  | 8.420849 | 18.92314 |
| DPS          | 96  | .2779562 | .9362707  | 0        | 8.754333 |
| NPL          | 96  | .0176199 | .0101005  | .0053939 | .0691208 |
| CIR          | 96  | .4475165 | .1305666  | .2270615 | .8695538 |
| ForeignOwn~p | 96  | .1913667 | .0969212  | .0001    | .3       |

### • Correlation coefficient matrix

```
. pwcorr STOCKPRICE GDP INF SIZE EPS BM DPS NPL CIR ForeignOwnership
```

|              | STOCKP~E       | GDP            | INF            | SIZE           | EPS            | BM             | DPS           |
|--------------|----------------|----------------|----------------|----------------|----------------|----------------|---------------|
| STOCKPRICE   | <b>1.0000</b>  |                |                |                |                |                |               |
| GDP          | <b>-0.1252</b> | <b>1.0000</b>  |                |                |                |                |               |
| INF          | <b>-0.0991</b> | <b>0.4263</b>  | <b>1.0000</b>  |                |                |                |               |
| SIZE         | <b>0.6023</b>  | <b>-0.0185</b> | <b>-0.0314</b> | <b>1.0000</b>  |                |                |               |
| EPS          | <b>0.7344</b>  | <b>-0.0019</b> | <b>0.0447</b>  | <b>0.6866</b>  | <b>1.0000</b>  |                |               |
| BM           | <b>-0.6825</b> | <b>0.1992</b>  | <b>0.2040</b>  | <b>-0.4206</b> | <b>-0.4365</b> | <b>1.0000</b>  |               |
| DPS          | <b>-0.0977</b> | <b>0.0132</b>  | <b>-0.0218</b> | <b>-0.1780</b> | <b>-0.2308</b> | <b>-0.0123</b> | <b>1.0000</b> |
| NPL          | <b>-0.4318</b> | <b>0.0854</b>  | <b>0.0884</b>  | <b>-0.3848</b> | <b>-0.5350</b> | <b>0.3899</b>  | <b>0.0652</b> |
| CIR          | <b>-0.4244</b> | <b>0.0893</b>  | <b>0.0626</b>  | <b>-0.6668</b> | <b>-0.6551</b> | <b>0.2914</b>  | <b>0.2290</b> |
| ForeignOwn~p | <b>0.2493</b>  | <b>-0.0233</b> | <b>-0.0034</b> | <b>0.0758</b>  | <b>0.2471</b>  | <b>-0.3113</b> | <b>0.0867</b> |

|              | NPL            | CIR           | Foreign~p     |
|--------------|----------------|---------------|---------------|
| NPL          | <b>1.0000</b>  |               |               |
| CIR          | <b>0.4895</b>  | <b>1.0000</b> |               |
| ForeignOwn~p | <b>-0.4346</b> | <b>0.0800</b> | <b>1.0000</b> |

- VIF index between variables

```
. vif
```

| Variable     | VIF         | 1/VIF           |
|--------------|-------------|-----------------|
| EPS          | <b>2.77</b> | <b>0.360845</b> |
| CIR          | <b>2.74</b> | <b>0.364756</b> |
| SIZE         | <b>2.37</b> | <b>0.421115</b> |
| NPL          | <b>1.96</b> | <b>0.510320</b> |
| ForeignOwn~p | <b>1.61</b> | <b>0.620707</b> |
| BM           | <b>1.52</b> | <b>0.658321</b> |
| INF          | <b>1.28</b> | <b>0.781636</b> |
| GDP          | <b>1.26</b> | <b>0.794024</b> |
| DPS          | <b>1.10</b> | <b>0.909678</b> |
| Mean VIF     | <b>1.85</b> |                 |

- OLS model

```
. regress STOCKPRICE GDP INF SIZE EPS BM DPS NPL CIR ForeignOwnership
```

| Source   | SS         | df | MS         | Number of obs | = | 96     |
|----------|------------|----|------------|---------------|---|--------|
|          |            |    |            | F(9, 86)      | = | 25.07  |
| Model    | 25154.2393 | 9  | 2794.91547 | Prob > F      | = | 0.0000 |
| Residual | 9587.63033 | 86 | 111.484074 | R-squared     | = | 0.7240 |
|          |            |    |            | Adj R-squared | = | 0.6952 |
| Total    | 34741.8696 | 95 | 365.70389  | Root MSE      | = | 10.559 |

| STOCKPRICE   | Coefficient | Std. err. | t     | P> t  | [95% conf. interval] |           |
|--------------|-------------|-----------|-------|-------|----------------------|-----------|
| GDP          | -52.83309   | 74.55138  | -0.71 | 0.480 | -201.0363            | 95.37015  |
| INF          | -29.96223   | 69.92605  | -0.43 | 0.669 | -168.9706            | 109.0462  |
| SIZE         | 3.57927     | 1.986851  | 1.80  | 0.075 | -.3704602            | 7.528999  |
| EPS          | .0076176    | .0012152  | 6.27  | 0.000 | .0052018             | .0100334  |
| BM           | -19.29588   | 3.233503  | -5.97 | 0.000 | -25.72387            | -12.86788 |
| DPS          | .4456981    | 1.213108  | 0.37  | 0.714 | -1.96588             | 2.857276  |
| NPL          | -27.03725   | 150.1346  | -0.18 | 0.858 | -325.4949            | 271.4204  |
| CIR          | 29.56525    | 13.73762  | 2.15  | 0.034 | 2.25576              | 56.87473  |
| ForeignOwn~p | -12.57978   | 14.18673  | -0.89 | 0.378 | -40.78206            | 15.62251  |
| _cons        | -53.93404   | 42.01063  | -1.28 | 0.203 | -137.4484            | 29.58034  |

- Fixed Effect Model

```
. xtreg STOCKPRICE GDP INF SIZE EPS BM DPS NPL CIR ForeignOwnership, fe
```

```
Fixed-effects (within) regression      Number of obs   =      96
Group variable: ID                    Number of groups =       8

R-squared:                            Obs per group:
    Within = 0.6840                      min =      12
    Between = 0.6823                     avg =     12.0
    Overall = 0.6786                     max =      12

                                F(9,79)      =     19.00
corr(u_i, Xb) = 0.1626              Prob > F      =     0.0000
```

| STOCKPRICE                              | Coefficient | Std. err.                         | t     | P> t  | [95% conf. interval] |                   |
|-----------------------------------------|-------------|-----------------------------------|-------|-------|----------------------|-------------------|
| GDP                                     | -65.4728    | 50.13661                          | -1.31 | 0.195 | -165.2672            | 34.32162          |
| INF                                     | -45.84877   | 47.6393                           | -0.96 | 0.339 | -140.6724            | 48.97489          |
| SIZE                                    | 1.911703    | 2.889484                          | 0.66  | 0.510 | -3.839671            | 7.663078          |
| EPS                                     | .0068974    | .0011264                          | 6.12  | 0.000 | .0046553             | .0091394          |
| BM                                      | -14.97674   | 3.462177                          | -4.33 | 0.000 | -21.86803            | -8.085449         |
| DPS                                     | -.5975112   | .8249566                          | -0.72 | 0.471 | -2.239546            | 1.044524          |
| NPL                                     | -99.63402   | 106.747                           | -0.93 | 0.353 | -312.1087            | 112.8406          |
| CIR                                     | 36.83885    | 12.61063                          | 2.92  | 0.005 | 11.73802             | 61.93969          |
| ForeignOwn~p                            | 17.38778    | 15.36935                          | 1.13  | 0.261 | -13.20415            | 47.9797           |
| _cons                                   | -28.58193   | 57.54614                          | -0.50 | 0.621 | -143.1247            | 85.9608           |
| sigma_u                                 | 9.4568057   |                                   |       |       |                      |                   |
| sigma_e                                 | 6.9823791   |                                   |       |       |                      |                   |
| rho                                     | .64718536   | (fraction of variance due to u_i) |       |       |                      |                   |
| F test that all u_i=0: F(7, 79) = 16.81 |             |                                   |       |       |                      | Prob > F = 0.0000 |

## • Random Effects Model

```
. xtreg STOCKPRICE GDP INF SIZE EPS BM DPS NPL CIR ForeignOwnership, re
```

```
Random-effects GLS regression           Number of obs   =       96
Group variable: ID                     Number of groups  =        8

R-squared:                             Obs per group:
    Within = 0.6436                      min =       12
    Between = 0.7893                     avg =      12.0
    Overall = 0.7240                     max =       12

                                Wald chi2(9)       =      225.63
corr(u_i, X) = 0 (assumed)             Prob > chi2        =      0.0000
```

| STOCKPRICE   | Coefficient                         | Std. err. | z     | P> z  | [95% conf. interval] |           |
|--------------|-------------------------------------|-----------|-------|-------|----------------------|-----------|
| GDP          | -52.83309                           | 74.55138  | -0.71 | 0.479 | -198.9511            | 93.28493  |
| INF          | -29.96223                           | 69.92605  | -0.43 | 0.668 | -167.0148            | 107.0903  |
| SIZE         | 3.57927                             | 1.986851  | 1.80  | 0.072 | -.3148875            | 7.473427  |
| EPS          | .0076176                            | .0012152  | 6.27  | 0.000 | .0052358             | .0099994  |
| BM           | -19.29588                           | 3.233503  | -5.97 | 0.000 | -25.63343            | -12.95833 |
| DPS          | .4456981                            | 1.213108  | 0.37  | 0.713 | -1.931949            | 2.823346  |
| NPL          | -27.03725                           | 150.1346  | -0.18 | 0.857 | -321.2956            | 267.2211  |
| CIR          | 29.56525                            | 13.73762  | 2.15  | 0.031 | 2.640004             | 56.49049  |
| ForeignOwn~p | -12.57978                           | 14.18673  | -0.89 | 0.375 | -40.38526            | 15.2257   |
| _cons        | -53.93404                           | 42.01063  | -1.28 | 0.199 | -136.2734            | 28.40529  |
| sigma_u      | 0                                   |           |       |       |                      |           |
| sigma_e      | 6.9823791                           |           |       |       |                      |           |
| rho          | 0 (fraction of variance due to u_i) |           |       |       |                      |           |

- Hausman test

```
. hausman fixed random, sigmamore
```

Note: the rank of the differenced variance matrix (7) does not equal the number of coefficients being tested (9); be sure this is what you expect, or there may be problems computing the test. Examine the output of your estimators for anything unexpected and possibly consider scaling your variables so that the coefficients are on a similar scale.

|              | Coefficients |               | (b-B)<br>Difference | sqrt(diag(V_b-V_B))<br>Std. err. |
|--------------|--------------|---------------|---------------------|----------------------------------|
|              | (b)<br>fixed | (B)<br>random |                     |                                  |
| GDP          | -65.4728     | -52.83309     | -12.63971           | 13.78697                         |
| INF          | -45.84877    | -29.96223     | -15.88654           | 17.32                            |
| SIZE         | 1.911703     | 3.57927       | -1.667566           | 3.891556                         |
| EPS          | .0068974     | .0076176      | -.0007202           | .0011935                         |
| BM           | -14.97674    | -19.29588     | 4.319137            | 4.117544                         |
| DPS          | -.5975112    | .4456981      | -1.043209           | .2908254                         |
| NPL          | -99.63402    | -27.03725     | -72.59677           | 59.29747                         |
| CIR          | 36.83885     | 29.56525      | 7.273609            | 13.2259                          |
| ForeignOwn~p | 17.38778     | -12.57978     | 29.96756            | 18.40896                         |

b = Consistent under  $H_0$  and  $H_a$ ; obtained from `xtreg`.  
 B = Inconsistent under  $H_a$ , efficient under  $H_0$ ; obtained from `xtreg`.

Test of  $H_0$ : Difference in coefficients not systematic

chi2(7) = (b-B)'[(V\_b-V\_B)^(-1)](b-B)  
 = 51.45

Prob > chi2 = 0.0000

(V\_b-V\_B is not positive definite)

## • Heteroscedasticity Test

```
. xttest3
```

Modified Wald test for groupwise heteroskedasticity  
 in fixed effect regression model

$H_0$ :  $\sigma(i)^2 = \sigma^2$  for all  $i$

chi2 (8) = 2019.22  
 Prob > chi2 = 0.0000

### • Testing for autocorrelation

```
. xtserial STOCKPRICE GDP INF SIZE EPS BM DGPA NPL CIR ForeignOwnership
```

Wooldridge test for autocorrelation in panel data

H0: no first-order autocorrelation

```
      F( 1,      7) =      17.411
      Prob > F =      0.0042
```

### • GLS model overcomes autocorrelation and heteroscedasticity

```
. xtglsl STOCKPRICE GDP INF SIZE EPS BM DPS NPL CIR ForeignOwnership, panels(hete
> ro) corr(ar1) force
```

Cross-sectional time-series FGLS regression

Coefficients: **generalized least squares**

Panels: **heteroskedastic**

Correlation: **common AR(1) coefficient for all panels (0.7550)**

```
Estimated covariances      =      8      Number of obs      =      96
Estimated autocorrelations =      1      Number of groups   =      8
Estimated coefficients     =     10      Time periods       =     12
                                Wald chi2(9)      =     146.84
                                Prob > chi2       =     0.0000
```

| STOCKPRICE    | Coefficient | Std. err. | z     | P> z  | [95% conf. interval] |           |
|---------------|-------------|-----------|-------|-------|----------------------|-----------|
| GDP           | -39.68256   | 24.72169  | -1.61 | 0.108 | -88.13618            | 8.771056  |
| INF           | 1.797887    | 22.36719  | 0.08  | 0.936 | -42.04099            | 45.63677  |
| SIZE          | 3.311536    | 1.810221  | 1.83  | 0.067 | -.2364318            | 6.859503  |
| EPS           | .0030573    | .0008039  | 3.80  | 0.000 | .0014818             | .0046329  |
| BM            | -16.47732   | 1.979144  | -8.33 | 0.000 | -20.35637            | -12.59827 |
| DPS           | -.0618306   | .360176   | -0.17 | 0.864 | -.7677626            | .6441013  |
| NPL           | -29.31728   | 61.98366  | -0.47 | 0.636 | -150.803             | 92.16845  |
| CIR           | 13.8965     | 8.368022  | 1.66  | 0.097 | -2.50452             | 30.29752  |
| ForeignOwne~p | 22.877      | 9.846368  | 2.32  | 0.020 | 3.578474             | 42.17553  |
| _cons         | -44.76002   | 36.24419  | -1.23 | 0.217 | -115.7973            | 26.2773   |

| APPENDIX AAAAAA: |      |             |       |       |      |       |       |        |       |      |                   |
|------------------|------|-------------|-------|-------|------|-------|-------|--------|-------|------|-------------------|
| Name             | YEAR | STOCK PRICE | BM    | GDP   | INF  | SIZE  | DPS   | EPS    | CIR   | NPL  | Foreign Ownership |
| VietComBank      | 2013 | 1.428       | 74.0% | 5.42% | 6.0% | 19.97 | 54.0% | 1.8806 | 40.3% | 2.7% | 19.7%             |
| VietComBank      | 2014 | 1.504       | 51.0% | 5.98% | 1.8% | 20.17 | 60.7% | 1.8457 | 39.6% | 2.3% | 20.8%             |
| VietComBank      | 2015 | 1.642       | 38.6% | 6.68% | 0.6% | 20.33 | 52.5% | 1.9940 | 39.2% | 1.8% | 20.9%             |
| VietComBank      | 2016 | 1.550       | 37.7% | 6.21% | 4.7% | 20.48 | 38.9% | 2.3415 | 40.0% | 1.5% | 20.8%             |
| VietComBank      | 2017 | 1.735       | 26.9% | 6.81% | 2.6% | 20.76 | 29.3% | 2.5269 | 40.4% | 1.1% | 20.7%             |
| VietComBank      | 2018 | 1.728       | 32.3% | 7.08% | 3.0% | 20.79 | 19.8% | 4.0596 | 34.7% | 1.0% | 20.7%             |
| VietComBank      | 2019 | 1.955       | 24.2% | 7.02% | 5.2% | 20.92 | 15.7% | 4.9958 | 34.6% | 0.8% | 23.9%             |
| VietComBank      | 2020 | 1.991       | 25.9% | 2.91% | 0.2% | 21.01 | 12.0% | 4.9749 | 32.7% | 0.6% | 23.7%             |
| VietComBank      | 2021 | 1.897       | 37.4% | 2.56% | 1.8% | 21.07 | 8.6%  | 5.9308 | 31.2% | 0.6% | 23.6%             |
| VietComBank      | 2022 | 1.903       | 35.8% | 8.12% | 4.6% | 21.32 | 11.2% | 6.3343 | 31.2% | 0.7% | 23.6%             |
| VietComBank      | 2023 | 1.905       | 36.8% | 5.05% | 3.6% | 21.33 | 0.0%  | 6.5071 | 32.4% | 1.0% | 23.4%             |
| VietComBank      | 2024 | 1.960       | 38.5% | 7.09% | 2.9% | 21.46 | 0.1%  | 6.0531 | 33.6% | 1.0% | 23.0%             |
| ACB              | 2013 | 1.193       | 87.0% | 5.42% | 6.0% | 18.93 | 77.7% | 0.8969 | 66.5% | 3.0% | 30.0%             |
| ACB              | 2014 | 1.188       | 89.8% | 5.98% | 1.8% | 19.01 | 66.9% | 1.0619 | 63.8% | 2.2% | 30.0%             |

|             |             |       |       |       |      |       |       |        |       |      |       |
|-------------|-------------|-------|-------|-------|------|-------|-------|--------|-------|------|-------|
| <b>ACB</b>  | <b>2015</b> | 1.297 | 72.1% | 6.68% | 0.6% | 19.12 | 61.0% | 1.1472 | 64.7% | 1.3% | 30.0% |
| <b>ACB</b>  | <b>2016</b> | 1.246 | 89.1% | 6.21% | 4.7% | 19.27 | 0.0%  | 1.4785 | 61.9% | 0.9% | 30.0% |
| <b>ACB</b>  | <b>2017</b> | 1.567 | 44.1% | 6.81% | 2.6% | 19.47 | 6.1%  | 2.1484 | 54.4% | 0.7% | 30.0% |
| <b>ACB</b>  | <b>2018</b> | 1.471 | 56.9% | 7.08% | 3.0% | 19.61 | 1.9%  | 4.1190 | 47.8% | 0.7% | 30.0% |
| <b>ACB</b>  | <b>2019</b> | 1.358 | 73.5% | 7.02% | 5.2% | 19.76 | 0.0%  | 3.6281 | 51.6% | 0.5% | 30.0% |
| <b>ACB</b>  | <b>2020</b> | 1.449 | 58.4% | 2.91% | 0.2% | 19.91 | 0.0%  | 3.5543 | 42.0% | 0.6% | 30.0% |
| <b>ACB</b>  | <b>2021</b> | 1.340 | 75.9% | 2.56% | 1.8% | 20.08 | 0.0%  | 3.5540 | 34.9% | 0.8% | 30.0% |
| <b>ACB</b>  | <b>2022</b> | 1.538 | 50.2% | 8.12% | 4.6% | 20.23 | 0.0%  | 4.0528 | 40.3% | 0.7% | 30.0% |
| <b>ACB</b>  | <b>2023</b> | 1.378 | 76.4% | 5.05% | 3.6% | 20.39 | 21.1% | 4.1309 | 33.2% | 1.2% | 30.0% |
| <b>ACB</b>  | <b>2024</b> | 1.412 | 72.4% | 7.09% | 2.9% | 20.58 | 23.1% | 3.7589 | 32.5% | 1.5% | 30.0% |
| <b>BIDV</b> | <b>2013</b> | 1.260 | 62.6% | 5.42% | 6.0% | 20.12 | 13.2% | 1.4410 | 38.7% | 2.4% | 0.0%  |
| <b>BIDV</b> | <b>2014</b> | 1.104 | 93.2% | 5.98% | 1.8% | 20.29 | 47.9% | 1.7735 | 39.4% | 2.0% | 0.6%  |
| <b>BIDV</b> | <b>2015</b> | 1.314 | 60.1% | 6.68% | 0.6% | 20.56 | 45.2% | 1.8652 | 44.9% | 1.7% | 2.0%  |
| <b>BIDV</b> | <b>2016</b> | 1.152 | 90.9% | 6.21% | 4.7% | 20.73 | 47.5% | 1.8117 | 44.5% | 2.0% | 1.3%  |
| <b>BIDV</b> | <b>2017</b> | 1.407 | 56.0% | 6.81% | 2.6% | 20.91 | 36.7% | 2.0316 | 39.7% | 1.6% | 2.3%  |
| <b>BIDV</b> | <b>2018</b> | 1.537 | 46.4% | 7.08% | 3.0% | 21.00 | 1.2%  | 2.2060 | 36.2% | 1.9% | 2.4%  |
| <b>BIDV</b> | <b>2019</b> | 1.664 | 41.8% | 7.02% | 5.2% | 21.12 | 56.7% | 2.1252 | 35.9% | 1.7% | 18.0% |

|                   |             |       |        |       |      |       |       |        |       |      |       |
|-------------------|-------------|-------|--------|-------|------|-------|-------|--------|-------|------|-------|
| <b>BIDV</b>       | <b>2020</b> | 1.680 | 41.3%  | 2.91% | 0.2% | 21.14 | 36.9% | 1.7960 | 35.4% | 1.8% | 17.4% |
| <b>BIDV</b>       | <b>2021</b> | 1.569 | 46.0%  | 2.56% | 1.8% | 21.29 | 12.5% | 2.1432 | 31.1% | 1.0% | 16.8% |
| <b>BIDV</b>       | <b>2022</b> | 1.587 | 50.9%  | 8.12% | 4.6% | 21.48 | 1.6%  | 3.6271 | 32.7% | 1.2% | 17.2% |
| <b>BIDV</b>       | <b>2023</b> | 1.637 | 49.7%  | 5.05% | 3.6% | 21.56 | 0.5%  | 3.8553 | 34.4% | 1.3% | 17.2% |
| <b>BIDV</b>       | <b>2024</b> | 1.575 | 55.9%  | 7.09% | 2.9% | 21.74 | 0.4%  | 3.7121 | 34.5% | 1.4% | 16.8% |
| <b>VIETINBANK</b> | <b>2013</b> | 1.210 | 89.6%  | 5.42% | 6.0% | 20.17 | 72.2% | 1.5599 | 45.5% | 1.0% | 28.6% |
| <b>VIETINBANK</b> | <b>2014</b> | 1.140 | 107.1% | 5.98% | 1.8% | 20.31 | 65.2% | 1.5383 | 46.6% | 1.1% | 28.7% |
| <b>VIETINBANK</b> | <b>2015</b> | 1.270 | 81.0%  | 6.68% | 0.6% | 20.47 | 65.1% | 1.5354 | 47.1% | 0.9% | 29.5% |
| <b>VIETINBANK</b> | <b>2016</b> | 1.178 | 107.6% | 6.21% | 4.7% | 20.67 | 0.0%  | 1.8169 | 48.7% | 1.0% | 29.8% |
| <b>VIETINBANK</b> | <b>2017</b> | 1.384 | 70.8%  | 6.81% | 2.6% | 20.81 | 69.9% | 2.0032 | 46.2% | 1.1% | 30.0% |
| <b>VIETINBANK</b> | <b>2018</b> | 1.286 | 93.7%  | 7.08% | 3.0% | 20.88 | 0.0%  | 1.4173 | 49.5% | 1.6% | 29.6% |
| <b>VIETINBANK</b> | <b>2019</b> | 1.320 | 99.4%  | 7.02% | 5.2% | 20.94 | 0.3%  | 2.5452 | 38.8% | 1.2% | 29.7% |
| <b>VIETINBANK</b> | <b>2020</b> | 1.538 | 66.4%  | 2.91% | 0.2% | 21.02 | 13.6% | 3.7023 | 35.4% | 0.9% | 28.9% |
| <b>VIETINBANK</b> | <b>2021</b> | 1.530 | 57.5%  | 2.56% | 1.8% | 21.15 | 0.1%  | 2.9580 | 32.3% | 1.3% | 25.9% |
| <b>VIETINBANK</b> | <b>2022</b> | 1.435 | 82.1%  | 8.12% | 4.6% | 21.32 | 22.6% | 3.5340 | 29.9% | 1.2% | 27.8% |
| <b>VIETINBANK</b> | <b>2023</b> | 1.433 | 86.5%  | 5.05% | 3.6% | 21.43 | 0.0%  | 3.7327 | 29.0% | 1.1% | 27.1% |
| <b>VIETINBANK</b> | <b>2024</b> | 1.577 | 73.2%  | 7.09% | 2.9% | 21.59 | 0.0%  | 4.7454 | 27.5% | 1.2% | 27.1% |

|                  |             |       |        |       |      |       |       |        |       |      |       |
|------------------|-------------|-------|--------|-------|------|-------|-------|--------|-------|------|-------|
| <b>MB BANK</b>   | <b>2013</b> | 1.104 | 106.0% | 5.42% | 6.0% | 19.01 | 46.5% | 2.0306 | 35.9% | 2.4% | 10.0% |
| <b>MB BANK</b>   | <b>2014</b> | 1.117 | 109.0% | 5.98% | 1.8% | 19.12 | 32.4% | 2.1589 | 37.5% | 2.7% | 10.0% |
| <b>MB BANK</b>   | <b>2015</b> | 1.164 | 99.2%  | 6.68% | 0.6% | 19.21 | 23.0% | 1.5701 | 39.3% | 1.6% | 10.0% |
| <b>MB BANK</b>   | <b>2016</b> | 1.146 | 110.9% | 6.21% | 4.7% | 19.36 | 0.8%  | 1.6836 | 42.4% | 1.3% | 20.0% |
| <b>MB BANK</b>   | <b>2017</b> | 1.405 | 64.2%  | 6.81% | 2.6% | 19.56 | 30.2% | 1.9226 | 43.3% | 1.2% | 20.0% |
| <b>MB BANK</b>   | <b>2018</b> | 1.289 | 81.3%  | 7.08% | 3.0% | 19.71 | 17.7% | 2.8651 | 44.7% | 1.3% | 20.0% |
| <b>MB BANK</b>   | <b>2019</b> | 1.318 | 82.5%  | 7.02% | 5.2% | 19.84 | 16.2% | 3.4694 | 39.4% | 1.2% | 20.0% |
| <b>MB BANK</b>   | <b>2020</b> | 1.362 | 78.5%  | 2.91% | 0.2% | 20.02 | 0.0%  | 3.1034 | 38.6% | 1.1% | 22.9% |
| <b>MB BANK</b>   | <b>2021</b> | 1.461 | 57.2%  | 2.56% | 1.8% | 20.22 | 0.0%  | 3.4993 | 33.5% | 0.9% | 23.2% |
| <b>MB BANK</b>   | <b>2022</b> | 1.233 | 102.7% | 8.12% | 4.6% | 20.41 | 0.0%  | 4.0042 | 32.5% | 1.1% | 21.2% |
| <b>MB BANK</b>   | <b>2023</b> | 1.271 | 99.5%  | 5.05% | 3.6% | 20.67 | 10.8% | 4.0379 | 31.5% | 1.6% | 23.2% |
| <b>MB BANK</b>   | <b>2024</b> | 1.400 | 87.9%  | 7.09% | 2.9% | 20.84 | 12.2% | 4.3253 | 30.7% | 1.6% | 23.2% |
| <b>SACOMBANK</b> | <b>2013</b> | 1.236 | 86.8%  | 5.42% | 6.0% | 18.90 | 65.6% | 1.9511 | 55.3% | 1.5% | 4.7%  |
| <b>SACOMBANK</b> | <b>2014</b> | 1.255 | 87.8%  | 5.98% | 1.8% | 19.06 | 1.6%  | 1.9312 | 54.1% | 1.2% | 5.9%  |
| <b>SACOMBANK</b> | <b>2015</b> | 1.117 | 93.5%  | 6.68% | 0.6% | 19.49 | 0.1%  | 0.3592 | 62.2% | 5.8% | 11.1% |
| <b>SACOMBANK</b> | <b>2016</b> | 0.975 | 130.2% | 6.21% | 4.7% | 19.62 | 0.6%  | 0.0491 | 87.0% | 6.9% | 8.5%  |
| <b>SACOMBANK</b> | <b>2017</b> | 1.109 | 100.3% | 6.81% | 2.6% | 19.72 | 0.1%  | 0.6551 | 73.3% | 4.7% | 9.4%  |

|                  |             |       |        |       |      |       |        |        |       |      |       |
|------------------|-------------|-------|--------|-------|------|-------|--------|--------|-------|------|-------|
| <b>SACOMBANK</b> | <b>2018</b> | 1.077 | 114.3% | 7.08% | 3.0% | 19.82 | 0.0%   | 0.9925 | 67.1% | 2.1% | 12.6% |
| <b>SACOMBANK</b> | <b>2019</b> | 1.002 | 147.5% | 7.02% | 5.2% | 19.93 | 0.0%   | 1.3611 | 63.3% | 1.9% | 14.2% |
| <b>SACOMBANK</b> | <b>2020</b> | 1.228 | 95.0%  | 2.91% | 0.2% | 20.02 | 0.0%   | 1.4870 | 63.1% | 1.7% | 9.4%  |
| <b>SACOMBANK</b> | <b>2021</b> | 1.498 | 57.7%  | 2.56% | 1.8% | 20.07 | 0.0%   | 1.8096 | 55.1% | 1.5% | 17.5% |
| <b>SACOMBANK</b> | <b>2022</b> | 1.352 | 91.1%  | 8.12% | 4.6% | 20.20 | 0.0%   | 2.6738 | 41.8% | 1.0% | 26.9% |
| <b>SACOMBANK</b> | <b>2023</b> | 1.446 | 86.8%  | 5.05% | 3.6% | 20.33 | 0.0%   | 4.0943 | 49.2% | 2.3% | 22.4% |
| <b>SACOMBANK</b> | <b>2024</b> | 1.567 | 79.0%  | 7.09% | 2.9% | 20.43 | 0.0%   | 5.3508 | 48.8% | 2.4% | 23.5% |
| <b>EXIMBANK</b>  | <b>2013</b> | 1.097 | 95.1%  | 5.42% | 6.0% | 18.95 | 253.4% | 0.5331 | 65.3% | 2.0% | 27.2% |
| <b>EXIMBANK</b>  | <b>2014</b> | 1.107 | 89.4%  | 5.98% | 1.8% | 18.90 | 875.4% | 0.0456 | 69.6% | 2.5% | 25.5% |
| <b>EXIMBANK</b>  | <b>2015</b> | 1.049 | 95.5%  | 6.68% | 0.6% | 18.64 | 0.2%   | 0.0325 | 60.7% | 1.9% | 26.2% |
| <b>EXIMBANK</b>  | <b>2016</b> | 0.954 | 121.5% | 6.21% | 4.7% | 18.67 | 0.0%   | 0.2513 | 60.4% | 2.9% | 28.0% |
| <b>EXIMBANK</b>  | <b>2017</b> | 1.102 | 91.6%  | 6.81% | 2.6% | 18.82 | 0.0%   | 0.6693 | 57.6% | 2.3% | 30.0% |
| <b>EXIMBANK</b>  | <b>2018</b> | 1.148 | 86.2%  | 7.08% | 3.0% | 18.84 | 0.0%   | 0.5373 | 65.2% | 1.8% | 30.0% |
| <b>EXIMBANK</b>  | <b>2019</b> | 1.250 | 72.0%  | 7.02% | 5.2% | 18.94 | 0.0%   | 0.7045 | 60.2% | 1.7% | 30.0% |
| <b>EXIMBANK</b>  | <b>2020</b> | 1.284 | 71.1%  | 2.91% | 0.2% | 18.89 | 0.0%   | 0.8705 | 54.9% | 2.5% | 30.0% |
| <b>EXIMBANK</b>  | <b>2021</b> | 1.528 | 42.9%  | 2.56% | 1.8% | 18.93 | 0.0%   | 0.7853 | 53.4% | 2.0% | 29.7% |
| <b>EXIMBANK</b>  | <b>2022</b> | 1.446 | 59.6%  | 8.12% | 4.6% | 19.04 | 0.0%   | 2.3960 | 47.7% | 1.8% | 18.9% |

|                 |             |       |        |       |      |       |       |        |       |      |       |
|-----------------|-------------|-------|--------|-------|------|-------|-------|--------|-------|------|-------|
| <b>EXIMBANK</b> | <b>2023</b> | 1.264 | 70.3%  | 5.05% | 3.6% | 19.12 | 0.0%  | 1.2437 | 47.9% | 2.7% | 1.8%  |
| <b>EXIMBANK</b> | <b>2024</b> | 1.286 | 69.8%  | 7.09% | 2.9% | 19.30 | 15.7% | 1.7860 | 39.7% | 2.5% | 2.5%  |
| <b>SHB</b>      | <b>2013</b> | 0.839 | 169.4% | 5.42% | 6.0% | 18.78 | 0.1%  | 0.9590 | 78.6% | 4.1% | 9.3%  |
| <b>SHB</b>      | <b>2014</b> | 0.908 | 146.0% | 5.98% | 1.8% | 18.95 | 76.3% | 0.8924 | 49.9% | 2.0% | 9.8%  |
| <b>SHB</b>      | <b>2015</b> | 0.813 | 182.7% | 6.68% | 0.6% | 19.14 | 0.3%  | 0.8387 | 52.8% | 1.7% | 11.3% |
| <b>SHB</b>      | <b>2016</b> | 0.663 | 257.0% | 6.21% | 4.7% | 19.27 | 0.0%  | 0.8158 | 50.5% | 1.9% | 8.9%  |
| <b>SHB</b>      | <b>2017</b> | 0.968 | 141.1% | 6.81% | 2.6% | 19.47 | 0.0%  | 1.3752 | 44.9% | 2.3% | 7.2%  |
| <b>SHB</b>      | <b>2018</b> | 0.857 | 188.5% | 7.08% | 3.0% | 19.59 | 0.0%  | 1.3900 | 47.8% | 2.4% | 9.5%  |
| <b>SHB</b>      | <b>2019</b> | 0.813 | 236.7% | 7.02% | 5.2% | 19.72 | 0.0%  | 2.0097 | 42.1% | 1.9% | 10.7% |
| <b>SHB</b>      | <b>2020</b> | 1.230 | 80.8%  | 2.91% | 0.2% | 19.84 | 0.0%  | 1.4893 | 35.2% | 1.8% | 4.4%  |
| <b>SHB</b>      | <b>2021</b> | 1.348 | 59.7%  | 2.56% | 1.8% | 20.04 | 0.0%  | 1.8775 | 24.3% | 1.7% | 3.3%  |
| <b>SHB</b>      | <b>2022</b> | 0.998 | 140.6% | 8.12% | 4.6% | 20.13 | 0.0%  | 2.5201 | 22.7% | 2.8% | 6.1%  |
| <b>SHB</b>      | <b>2023</b> | 1.033 | 128.2% | 5.05% | 3.6% | 20.26 | 0.0%  | 2.0240 | 23.7% | 3.0% | 5.0%  |
| <b>SHB</b>      | <b>2024</b> | 1.011 | 154.7% | 7.09% | 2.9% | 20.43 | 18.7% | 2.5453 | 24.4% | 2.9% | 2.8%  |

